

BizFinBench.v2: Towards Reliable LLMs in Finance via Real-User Data and Offline/Online Bilingual Evaluation

Xin Guo^{1 2 *} Rongjunchen Zhang^{1 * †} Guilong Lu¹ Xuntao Guo¹ Shuai Jia¹ Zhi Yang² Liwen Zhang^{2 †}

Abstract

Large language models are becoming increasingly significant in financial applications. Nevertheless, prevailing benchmarks are largely dependent on simulated or generic data, which leads to a significant gap between reported performance and actual efficacy in real-world scenarios. To tackle this challenge, we present BizFinBench.v2, the first integrated offline and online benchmark built upon authentic user query-response data from both Chinese and U.S. equity markets. It comprises 28,860 questions across eight offline and two online tasks. Experimental results show that GPT-5 achieves a mere 61.5% accuracy, still failing to meet the practical business requirement (84.8%). Among the evaluated commercial models, DeepSeek-R1 exhibits superior investment efficacy. Error analysis grounded in real financial practice reveals persistent limitations in existing models. By overcoming the constraints of prior benchmarks, BizFinBench.v2 provides a substantiated foundation for advancing LLM deployment in the financial sector. Our data and code are available at <https://github.com/HiThink-Research/BizFinBench.v2>.

1. Introduction

Large Language Models (LLMs) have developed rapidly in recent years, and their application boundaries in the financial sector continue to expand, making them an important technological direction for promoting the intelligent upgrade of financial services (Chen et al., 2024; Liu et al., 2025a; Zhang et al., 2023b; Lu et al., 2024). However, as the

^{*}Equal contribution ¹HiThink Research ²Shanghai University of Finance and Economics. Correspondence to: Rongjunchen Zhang <zhangrongjunchen@myhexin.com>, Liwen Zhang <zhang.liwen@shufe.edu.cn>.

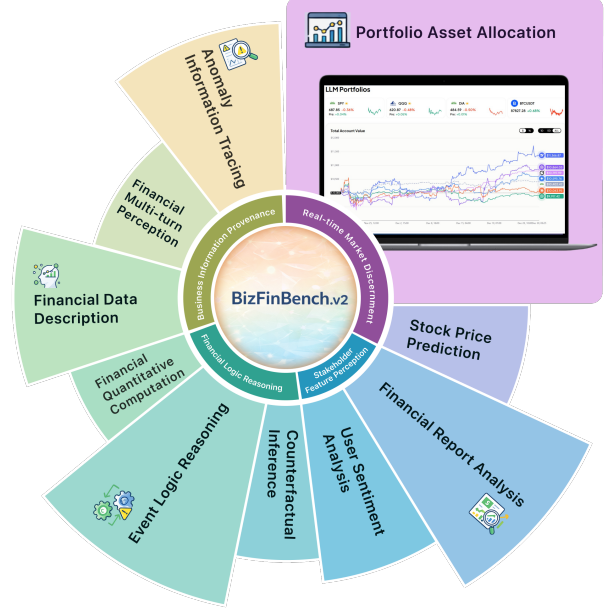


Figure 1. BizFinBench.v2 comprises eight foundational tasks and two online tasks distributed across four major scenarios. The top-right corner displays a real-time screenshot of the Portfolio Asset Allocation task.

scope of LLMs’ application in the financial sector becomes increasingly broad, the discrepancy between model performance under existing evaluation paradigms and their actual performance in real financial business scenarios has become increasingly prominent, directly leading to a conflict between real financial business needs and the limitations of various evaluation benchmarks.

The core characteristics of financial scenarios lie in authenticity and online capability, yet the vast majority of current financial benchmarks are undermined by two fundamental limitations:

- (1) **Disconnected from Real-world Business Scenarios:** Most existing studies rely on simulated samples or general datasets that are disconnected from actual business operations. These datasets generally exhibit low difficulty and are decoupled from the requirements of real-world financial business, as they lack the core logic of business workflows

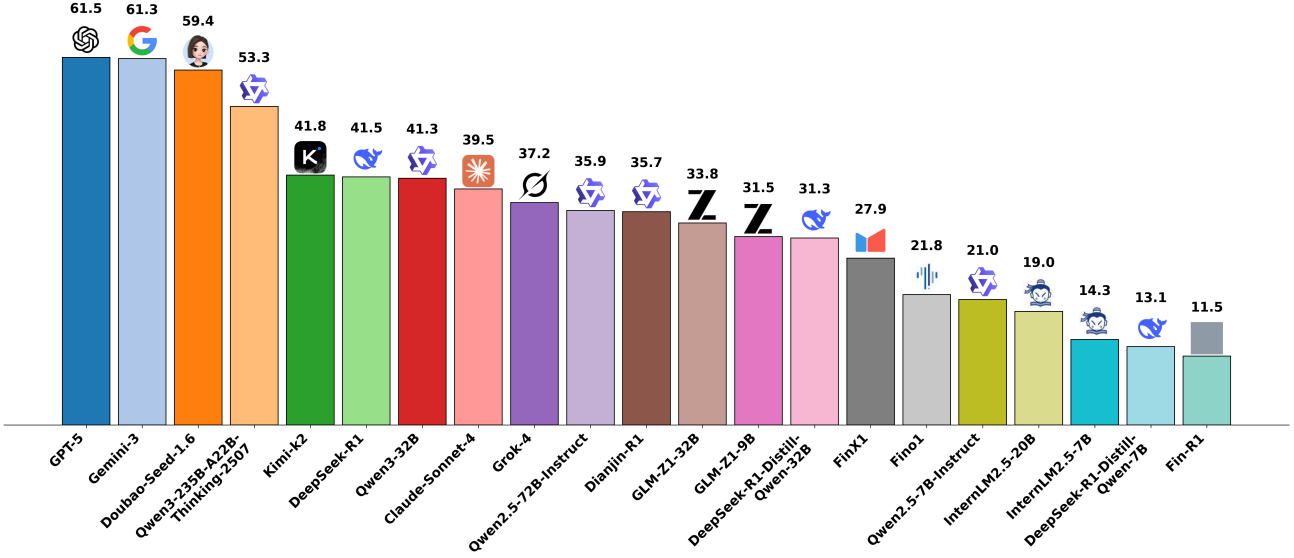


Figure 2. We have ranked the performance of the LLMs participating in the evaluation under the zero-shot setting, and these results reflect their authentic practical business capabilities.

and fail to reflect the practical challenges inherent in the financial domain. (Zhu et al., 2024; Jiang et al., 2025).

(2) Focusing on Purely Static Offline Tasks: Existing benchmarks universally focus on purely offline static tasks (Xie et al., 2024), lacking coverage of online tasks in financial scenarios. This prevents the support for evaluating model performance in online business scenarios such as real-time market data push and dynamic risk monitoring, thus being disconnected from the core requirement of online capability in financial services (Tang et al., 2025). These two problems result in a disconnect between existing benchmarks and real-world applications, leading to a significant gap between LLM benchmark test performance and actual business performance.

To bridge the gap between current LLM evaluation performance and actual operational efficacy, we introduce BizFinBench.v2, the first large-scale financial business evaluation benchmark that integrates authentic business data from both Chinese and U.S. equity markets and features a dual-track evaluation of “Core Business Capabilities + Online Performance”. BizFinBench.v2 comprises 28,860 QA pairs based on real business data. Through the clustering of massive user queries from authentic business platforms and rigorous quality control, we have categorized four core business scenarios: Business Information Provenance, Financial Logic Reasoning, Stakeholder Feature Perception, and Real-time Market Discernment. These encompass eight fundamental tasks, including Anomaly Information Tracing, Financial Multi-turn Perception, Financial Data Description, Financial Quantitative Computation, Event Logic Reasoning, Coun-

terfactual Inference, User Sentiment Analysis and Financial Report Analysis, along with two online tasks: Stock Price Prediction and Portfolio Asset Allocation. The detailed architecture is illustrated in Figure 1.

Unlike existing benchmarks, BizFinBench.v2 not only enables a precise and objective quantitative evaluation of the actual efficacy of LLMs in authentic financial scenarios but also identifies core capability shortcomings in financial business applications. It achieves a comprehensive deconstruction of LLM capabilities from a business perspective, thereby providing a reliable performance reference for the deployment of LLMs in the financial domain.

Our contributions are threefold:

(1) Authentic Business Data Driven: We propose the first large-scale financial evaluation benchmark driven by authentic data. Developed upon a core foundation of real-world business data from Chinese and U.S. equity markets, BizFinBench.v2 transcends the limitations inherent in existing evaluation frameworks, specifically their disconnection from practical business scenarios and their reliance on low-complexity, general-purpose, or synthetic data.

(2) Innovative Dual-Track Evaluation Framework: We propose a dual-track evaluation framework encompassing “Core Business Capabilities + Online Performance”, thereby addressing the deficiencies in application scenario coverage inherent in traditional offline static assessments.

(3) Expert-Informed Business Error Analysis: We provide an error analysis of the actual performance of LLMs from the business perspective of financial experts, offer-

ing targeted optimization directions for broader business adaptation.

The organization of this paper is as follows: Section 2 deliberates on the current status of various operations and benchmarks in the financial domain, identifying the corresponding requirements and the limitations of existing research. Section 3 delineates the construction, quality control, and detailed statistics of BizFinBench.v2. Sections 4 and Section 5 primarily describe the experimental setup, as well as the main experimental results and analysis. Section 6 presents an error analysis conducted from an expert business perspective. Finally, Section 7 concludes the study and proposes directions for future research.

2. Related Work

2.1. Financial Business Analysis

In financial business scenarios, LLM-based tasks such as text understanding (Wilson et al., 2024), sentiment analysis (Delgadillo et al., 2024; Zhang et al., 2023a), time-series forecasting (Li et al., 2024a;b; Wang et al., 2024; Mai, 2024), and wealth management & investment (Yu et al., 2024; Wang et al., 2025; Yu et al., 2025; Yang et al., 2023; Liu et al., 2025c) serve as foundational components for building universal financial business capabilities. They provide crucial technical support for judging market trends and mitigating potential risks. However, the financial sector’s demand for timeliness means standalone offline tasks cannot fully evaluate LLMs’ performance in real-world business contexts—especially scenarios closely tied to dynamic financial markets, such as stock price fluctuation prediction (Yu & Yan, 2020), real-time response to market hot topics and public opinion (Mao et al., 2025; Yang et al., 2026), intraday trading signal interpretation (Cervelló-Royo et al., 2015), and instant investment decisions (Suresh, 2024; Han et al., 2026). Nevertheless, most current research on LLMs in finance focuses on enhancing offline question-answering quality and optimizing reasoning accuracy, verifying model performance through static datasets (Liu et al., 2025b; Zhu et al., 2025; Hou et al., 2026). Few studies test LLMs’ capabilities in processing real-time market data and generating instantaneous decisions from the perspective of financial businesses’ online and real-time needs. This gap results in a mismatch between LLM performance and the core demands of financial markets, leaving significant room for future exploration.

2.2. Financial Benchmark Analysis

Mainstream evaluation benchmarks in the current financial domain have achieved multi-dimensional coverage for assessing the foundational capabilities of LLMs. Benchmarks such as CFLUE (Zhu et al., 2024), FinEval (Guo et al.,

2025b), and CGCE (Zhang et al., 2023c) focus on basic financial knowledge and text comprehension, encompassing tasks like text classification and policy interpretation. Others including FinQA (Chen et al., 2021), ConvFinQA (Chen et al., 2022), FinanceReasoning (Tang et al., 2025), and FinBen (Xie et al., 2024) emphasize financial numerical reasoning and complex decision-making, simulating quantitative analysis scenarios such as financial report interpretation. Meanwhile, benchmarks like DISC-FINSFT (Chen et al., 2023), PIXIU (Xie et al., 2023), FinMaster (Jiang et al., 2025) and CFBenchmark (Lei et al., 2023) prioritize instruction following and tool adaptation. However, these benchmarks are generally focused on verifying performance in single, offline, and static tasks, failing to accommodate the real-time demands inherent in financial services. Furthermore, the construction of the relevant data is mostly derived from simulated synthesis or manual rewriting, making it difficult to align with real-world financial business requirements. Simultaneously, there is a lack of in-depth discussion on the consistency between evaluated performance and the LLM’s actual financial application capability, leading to a divergence between the practical guidance provided by the benchmarks and the core demands of the financial market. Detailed comparison with existing benchmarks can be found in Table 1.

Table 1. In-depth multi-dimensional comparison between BizFinBench.v2 and existing benchmarks from a real-business perspective. The abbreviations and their core meanings are as follows: Cross-Lingual (CL), Online Testing (OT), Real Business Data (RBD).

Benchmarks	CL	OT	RBD
FinEval (Guo et al., 2025b)	✗	✗	✗
CFLUE (Zhu et al., 2024)	✓	✗	✗
FinQA (Chen et al., 2021)	✗	✗	✗
ConvFinQA (Chen et al., 2022)	✗	✗	✗
FinMaster (Jiang et al., 2025)	✗	✗	✗
FinBen (Xie et al., 2024)	✓	✗	✗
FinanceMath (Zhao et al., 2024)	✗	✗	✗
FinanceReasoning (Tang et al., 2025)	✗	✗	✗
BizFinBench.v2	✓	✓	✓

3. BizFinBench.v2 Benchmark

3.1. Overview

As the first large-scale financial evaluation benchmark to cover authentic business data from both Chinese and U.S. equity markets while integrating offline evaluation and online testing, BizFinBench.v2 aims to bridge the gap between the performance of LLMs in the financial domain and their actual business capabilities. All data within the benchmark is derived from real financial business platforms and has undergone rigorous compliance review and quality control. All data will be released in the future for research purposes.

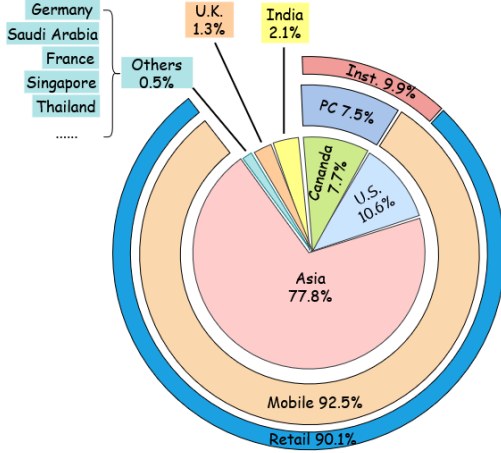


Figure 3. The platform’s user structure is primarily analyzed through user type, device distribution, and national distribution; here, Inst. serves as an abbreviation for institutional users.

3.2. Task Construction

In the process of task construction, BizFinBench.v2 anchors itself in the core functions of authentic financial business platforms and is guided by the actual demands of key market participants. It covers the diverse needs of individual users, including novices, short-term investors, and high-net-worth individuals, while accounting for the critical requirements of mainstream financial institutions such as banks and securities firms. The platform user distribution is shown in Figure 3.

Business Information Provenance (BIP) focuses on frontline scenarios, requiring LLMs to resolve various business inquiries through direct interaction. This includes real-world perturbations such as erroneous input, irrelevant information, or typos, necessitating robust discernment capabilities from the models. Specifically, the Anomaly Information Tracing (AIT) task addresses user inquiries regarding the root causes of market anomalies, such as significant fluctuations in watchlist stocks; it requires LLMs to filter key clues from massive multi-dimensional information, exclude interference, and precisely identify the core events driving stock or market volatility. The Financial Multi-turn Perception (FMP) task requires LLMs to respond to continuous inquiries within long-context interactions, integrating historical user queries to provide sustained service. The Financial Data Description (FDD) task demands that LLMs judge the authenticity and accuracy of specific data within various information sources to provide precise conclusions.

The Financial Logical Reasoning (FLR) scenario serves as both a foundational element of Business Information Provenance, requiring LLMs to perform rigorous financial reasoning based on market data returned by various interfaces. Within this category, the Financial Quantitative

Computation (FQC) task requires LLMs complete accurate indicator calculations based on complex financial data. The Event Logical Reasoning (ELR) task requires LLMs to sort macro or micro market events based on chronological order or causal logic. The Counterfactual Inference (CI) task requires LLMs to simulate financial experts by conducting reasonable logical analysis and judgment based on user-proposed hypotheses, outputting reliable and valid conclusions.

The Stakeholder Feature Perception (SFP) scenario aims to provide users with in-depth analysis and summaries of the market or industry. The User Sentiment Analysis (SA) task requires LLMs to classify and evaluate overall user sentiment through the quantitative analysis of various user information and behaviors, providing support for customized services and highly relevant product recommendations. The Financial Report Analysis (FRA) task requires LLMs to conduct quantitative analysis by combining technical, fundamental, and news-based periodic report information to determine enterprise industry rankings, helping users intuitively grasp corporate qualifications and future prospects. This task is limited to the Chinese market.

Finally, the Real-time Market Discernment (RMD) scenario covers two types of online tasks of high interest to users: Stock Price Prediction (SPP) and Portfolio Asset Allocation (PAA). It should be noted that the real-time nature of online tasks requires all relevant data to be derived from the actual equity market, and theoretically, the number of questions is unlimited. However, constrained by the static presentation format of the paper, we set the cutoff time for the SPP task to December 24, 2025, and incorporated its evaluation results into the offline benchmark for more direct comparison. The Stock Price Prediction task addresses user needs for predicting stocks in their watchlists, requiring LLMs to predict daily closing prices by integrating historical and real-time stock prices, technical indicators, and relevant news. The Portfolio Asset Allocation task presents a higher degree of difficulty. We have constructed an LLM investment simulation system for this task that restores real-world trading rules, including transaction fees, latency, and slippage, on top of real-time data access. For detailed settings, refer to Section 4.2. Through systematic prompt engineering and various data and tool interfaces, LLMs can perform asset allocation in a real market environment using simulated funds. Decisions are generated hourly during the day, allowing for either trading or skipping, with each decision cycle covering the full process from reasoning and analysis to order submission and trade execution. We will subsequently open-source this LLM investment system to allow all researchers interested in the financial field to integrate different open-source or proprietary LLMs for in-depth study. For a more detailed introduction to the above content, please refer to Appendix A.

3.3. Quality Control

It should be emphasized here that all questions and corresponding answers in BizFinBench.v2 are sourced from authentic user platforms. The quality control process is primarily designed to filter out low-quality queries and perform normalization to ensure suitability for LLM evaluation. The specific implementation details are as follows.

For tasks in offline tasks, the quality control process strictly adopts a three-level progressive mechanism of “Platform Clustering and Desensitization - Frontline Staff Review - Expert Team Cross-Validation” to ensure the validity and compliance of the evaluation data. First, all raw data undergo clustering through the platform’s multi-dimensional algorithms to automatically aggregate and categorize different task types. Simultaneously, data desensitization is performed (including user PII or sensitive corporate data), and a preliminary review is conducted in alignment with financial industry compliance requirements to eliminate content that clearly violates regulatory standards. Second, ten frontline business professionals with over five years of experience manually screen the data samples one by one to filter out invalid Q&A pairs, duplicate records, and samples with abnormal expressions, while retaining specific real-world perturbations to ensure each Q&A pair fully reflects authentic business scenarios. Finally, we invited six senior financial experts, each with over a decade of industry experience from three different professional teams within the Financial Research Institute, to conduct cross-validation in three groups. Ultimately, through this rigorous three-level screening mechanism, high-quality Q&A pairs are formed that balance authenticity, business alignment, and information security.

As for online tasks, the financial experts mentioned above strictly defined the structured data required for the Stock Price Prediction task, the specific equity market configurations for the Portfolio Asset Allocation task, and the system prompts, ensuring full compliance with the requirements of authentic business scenarios.

It should be noted that for offline tasks, due to the protection of financial business privacy, the specific evaluation rubrics and quantitative standards employed during the quality review process cannot be disclosed to the public. In contrast, for online tasks, as real-time public equity market data is utilized and the configurations are designed to replicate the actual market environment, the details of these settings and the system prompts will be made open-source.

3.4. Statistics

BizFinBench.v2 covers the two major equity markets (China and the U.S.), with all data drawn from authentic user demands within financial business scenarios. This has resulted

in four core scenarios: BIP, FLR, SFP and RMD. These comprise eight offline tasks and two online tasks, totaling 28,860 questions derived from real-world financial service requests. Specifically, the BIP scenario comprises a total of 12,263 questions, including 3,963 AIT tasks, 4,497 FMP tasks, and 3,803 FDD tasks; the average input token counts for these three tasks are 8,679, 10,361, and 3,577, respectively. The FLR scenario consists of 6,548 questions, containing 2,000 FQC tasks, 3,944 ELR tasks, and 604 CI tasks, with corresponding average input token counts of 1,984, 437, and 2,267. The SFR scenario includes 6,000 questions, divided into 4,000 SA tasks and 2,000 FRA tasks, with average input token counts of 3,326 and 19,681, respectively. In the RMD scenario, both SPP and PAA tasks are real-time tasks. Since the SPP task utilizes the same accuracy metric as offline tasks, we report the 4,049 questions generated during the evaluation process. Conversely, as the PAA task primarily employs long-term outcomes as evaluation metrics, the total number of its transactions is not included in the aggregate data. It should be emphasized that the number of questions for this task depends on the time periods and trading frequencies (e.g., second-level, minute-level, etc.) defined by researchers; thus, there is theoretically no fixed upper limit. More intuitive statistical results can be found in Table 5 within Appendix A.

4. Experiment Setting

4.1. Evaluated Methods

All questions in BizFinBench.v2 are verifiable open-ended questions. We provide several supplementary notes here: (1) Regarding SA and SPP tasks, we require the LLM to output a prediction interval based on its predicted value, consistent with the error tolerance of actual business operations. Correctness is determined by verifying whether the provided prediction interval contains the ground truth. Specifically, the business tolerance for the SA task is set at 10%, while the tolerance for the SPP task is set at 1%. (2) The evaluation metrics for the PAA task primarily focus on investment indicators such as cumulative return, Sharpe ratio, and maximum drawdown. Therefore, the models evaluated for this task (mainly the six currently most powerful proprietary commercial models) and their corresponding experimental results are independent of other tasks.

Therefore, for tasks using average accuracy as the metric, we evaluate model performance under both zero-shot and Chain-of-thought (CoT) settings. For the PAA task, we only provide the final indicators.

4.2. Evaluated Models

We evaluated 21 LLMs, where proprietary models were accessed via their respective APIs and open-source models

Table 2. LLM performance in zero-shot setting on BizFinBench.v2 (%). The results are color-coded to indicate the top three performers in each task: **gold** represents the best-performing model, **silver** represents the second-best result, and **bronze** represents the third-best performance. The meanings of the abbreviations are as follows: AIT(Anomaly Information Tracing), FMP(Financial Multi-turn Perception), FDD(Financial Data Description), FQC(Financial Quantitative Computation), ELR(Event Logic Reasoning), CI(Counterfactual Inference), SA(User Sentiment Analysis), FRA(Financial Report Analysis), SPP(Stock Price Prediction).

Model	Size	AIT	FMP	FDD	FQC	ELR	CI	SA	FRA	SPP	Average
Proprietary LLMs											
GPT-5	unknown	54.2	90.8	68.3	89.2	62.0	83.9	18.8	54.1	32.1	61.5
Gemini-3	unknown	64.8	87.0	69.7	85.8	69.5	82.2	7.4	50.8	34.9	61.3
Doubao-Seed-1.6	unknown	62.6	90.2	63.8	78.2	61.2	78.7	22.7	46.3	31.1	59.4
Kimi-k2	unknown	55.2	80.9	22.4	62.2	44.6	15.4	20.1	45.0	30.5	41.8
Claude-Sonnet-4	unknown	54.8	79.9	28.4	29.8	44.8	23.4	17.3	47.7	29.1	39.5
Grok-4	unknown	61.8	86.6	37.4	9.3	42.1	4.7	17.8	45.2	30.3	37.2
Open-source General LLMs											
Qwen3-235B-A22B-Thinking-2507	235B	49.3	87.9	68.0	76.0	50.6	72.2	16.4	22.5	36.9	53.3
DeepSeek-R1	671B	58.9	87.2	42.1	21.7	48.9	8.1	23.9	50.0	32.3	41.5
Qwen3-32B	32B	54.3	80.9	40.0	48.4	42.0	47.2	13.4	40.5	5.1	41.3
Qwen2.5-72B-Instruct	72B	61.0	78.2	26.5	19.8	39.9	20.7	21.6	47.0	8.2	35.9
GLM-Z1-32B	32B	49.6	66.9	45.6	34.4	31.6	31.3	1.2	36.8	6.8	33.8
GLM-Z1-9B	9B	46.5	69.2	40.4	26.9	35.2	25.2	0.4	36.7	3.4	31.5
DeepSeek-R1-Distill-Qwen-32B	32B	52.1	75.4	20.5	21.0	35.9	24.2	18.4	27.3	6.8	31.3
Qwen2.5-7B-Instruct	7B	35.2	43.2	33.8	0.8	20.9	1.0	23.3	28.3	2.8	21.0
InternLM2.5-20B	20B	32.1	41.1	32.0	0.2	27.8	0.7	3.7	32.5	0.6	19.0
InternLM2.5-7B	7B	28.3	14.8	30.6	0.5	18.9	0.3	6.7	28.5	0.0	14.3
DeepSeek-R1-Distill-Qwen-7B	7B	16.6	30.9	17.1	3.5	10.3	6.2	19.1	13.0	1.0	13.1
Open-source Financial LLMs											
Dianjin-R1	32B	54.2	70.7	40.9	25.3	45.8	22.0	6.7	46.0	9.9	35.7
FinX1	70B	47.9	73.0	29.5	14.3	31.5	11.6	5.3	34.7	3.6	27.9
Fino1	14B	27.5	38.6	24.3	14.1	14.6	22.1	11.1	35.3	8.6	21.8
Fin-R1	7B	21.2	29.1	0.5	1.5	9.8	3.2	10.9	24.5	2.9	11.5
Financial Experts	–	92.6	98.0	94.5	100	91.7	100	57.9	96.0	32.3	84.8

were deployed locally. All inference tasks were executed on an $8 \times$ NVIDIA H200 GPU cluster. For additional details regarding the models, refer to Appendix B.1. We additionally invited two financial experts who were not involved in the data construction process to participate in the competition, so as to conduct a comparative analysis with the performance of LLMs.

In the realm of proprietary LLMs, we evaluated 6 leading, high-performance models, including GPT-5 (Achiam et al., 2023), Gemini-3 (Team et al., 2023), Claude-Sonnet-4 (Anthropic, 2025), Grok-4 (xAI, 2025), Doubao-Seed-1.6 (Bytedance, 2025) and Kimi-k2 (Team et al., 2025). For open-source models, we evaluated 11 models from several mainstream LLMs, including Qwen2.5-7B/72B-Instruct, Qwen3-32B and Qwen3-235B-A22B-Thinking-2507 (Yang et al., 2025), InternLM2.5-7B/20B (Team, 2023), GLM-Z1-9B/32B (Glm et al., 2024), DeepSeek-R1 and DeepSeek-R1-Distill-Qwen-7B/32B (Guo et al., 2025a). In financial domain, we evaluated 4 representative LLMs tailored for the financial tasks, including Fin-R1 (Liu et al., 2025b), Dianjin-R1 (Zhu et al., 2025), FinX1 (Team, 2024) and Fino1 (Qian et al., 2025).

In the PAA task, we primarily evaluated six of the most capable commercial models currently available on the market, namely GPT-5, Gemini-3, Grok-4, Claude-Sonnet-4, DeepSeek-R1, and Qwen3-Max (Yang et al., 2025). To maintain consistency, DeepSeek-R1 was also accessed via API. Simultaneously, we introduce the moving average crossover strategy, the equal-weight allocation strategy, and the basic momentum strategy as traditional quantitative baselines for comparative experiments. Specifically, the moving average crossover strategy utilizes an MA5-MA20 combination; the equal-weight allocation strategy is configured with three rebalancing frequencies: weekly, monthly, and quarterly; and the basic momentum strategy adopts top-5 and top-10 stock selection grouping methods. Finally, specific configurations for relevant parameters in the PAA task (including slippage, spreads, and liquidity, among others) are detailed in Appendix E.

5. Main Results

The ranking results of the models under the zero-shot setting are shown in Figure 2, and Table 2 provides the complete

Table 3. LLM performance in CoT settings on BizFinBench.v2 (%). Other relevant settings and abbreviations are identical to those in Table 2.

Model	Size	AIT	FMP	FDD	FQC	ELR	CI	SA	FRA	SPP	Average
Proprietary LLMs											
Doubao-Seed-1.6	unknown	64.3	88.1	61.2	79.2	60.0	75.7	20.3	47.8	6.7	56.9
Gemini-3	unknown	67.8	91.8	54.9	84.6	67.9	80.6	3.7	51.0	0.9	56.5
GPT-5	unknown	54.1	88.6	49.3	86.7	59.5	83.6	8.8	55.7	6.0	54.6
Kimi-k2	unknown	63.7	81.4	27.2	40.8	45.0	42.4	17.8	48.3	2.6	40.1
Grok-4	unknown	63.2	88.7	39.2	20.5	43.5	20.2	18.3	44.7	7.7	37.7
Claude-Sonnet-4	unknown	21.8	39.3	13.4	9.0	13.1	8.1	2.9	48.3	1.6	13.7
Open-source General LLMs											
Qwen3-235B-A22B-Thinking-2507	235B	48.4	88.5	63.6	75.3	58.4	70.6	12.8	21.7	4.2	52.7
DeepSeek-R1	671B	61.7	87.2	43.2	54.2	50.9	55.9	21.6	46.5	32.8	50.4
Qwen3-32B	32B	55.1	79.7	12.9	44.7	41.4	43.9	9.3	39.8	3.8	36.4
Qwen2.5-72B-Instruct	72B	60.9	82.0	31.0	28.8	39.6	27.2	13.0	45.2	7.6	36.3
DeepSeek-R1-Distill-Qwen-32B	32B	54.4	77.3	22.1	19.7	40.6	20.2	17.8	27.3	6.7	32.4
GLM-Z1-32B	32B	50.9	62.4	42.8	32.7	28.8	30.6	0.8	35.7	6.8	32.0
GLM-Z1-9B	9B	45.9	72.9	41.3	24.8	32.7	25.2	0.6	36.2	3.4	30.9
Qwen2.5-7B-Instruct	7B	26.1	49.4	31.9	4.9	21.1	5.0	16.3	28.5	2.5	19.6
InternLM2.5-20B	20B	33.0	58.1	17.4	1.7	29.4	2.0	3.0	33.7	0.4	18.1
DeepSeek-R1-Distill-Qwen-7B	7B	14.4	30.3	16.0	3.0	10.6	3.8	6.5	13.0	0.9	10.7
InternLM2.5-7B	7B	26.0	23.8	4.8	1.7	19.3	0.7	5.8	27.0	0.1	10.2
Open-source Financial LLMs											
Dianjin-R1	32B	53.0	76.6	39.7	18.0	41.3	20.0	5.6	30.0	5.8	32.5
FinX1	70B	45.8	74.4	25.3	8.5	32.0	7.1	2.5	34.7	3.0	24.8
Finol	14B	36.5	58.4	0.0	13.5	28.2	8.9	12.8	28.5	8.8	20.9
Fin-R1	7B	22.7	30.8	35.7	1.8	10.7	3.6	7.5	24.2	0.6	14.2

Table 4. Comparison of key metrics between commercial LLMs and traditional quantitative strategies over the same period in Portfolio Asset Allocation tasks. The primary indicators include Total Return (TR), Profit Factor (PF), Sharpe Ratio (SR), Maximum Drawdown (MD), and Total Assets (TA). The abbreviation EW_week stands for EqualWeight_week, M.Top5 stands for Momentum_Top5, and other abbreviations follow the same pattern.

Model	TR	PF	SR	MD	TA
LLM					
DeepSeek-R1	+47.34%	1.80	1.25	-53%	14734
Claude-Sonnet-4	-2.11%	0.90	-0.05	-11%	9789
Qwen3-Max	-3.79%	0.85	-0.10	-13%	9621
Gemini-3	-6.61%	0.83	-0.20	-13%	9339
Grok-4	-10.15%	0.76	-0.32	-15%	8985
GPT-5	-13.80%	0.72	-0.45	-16%	8620
Quantitative Strategy					
MA5-MA20	-1.61%	1.13	-3.97	-2.21%	9839
EW_week	22.41%	4.74	2.21	-3.05%	12241
EW_month	22.08%	3.34	1.68	-6.39%	12208
EW_quarter	32.98%	1.96	2.19	-9.20%	13298
M.Top5	101.71%	2.38	1.60	-56.89%	20171
M.Top10	78.85%	1.39	1.65	-41.51%	17885
Baseline					
SPY	-	-	-	-	10632

quantitative results. Overall, GPT-5 ranks first with an average accuracy of 61.5%, highlighting its comprehensive competitive advantage; proprietary models such as Gemini-3 and Doubao-Seed-1.6 performed outstandingly, ranking among the top three in multiple tasks. Among open-source models, Qwen3-235B-A22B-Thinking-2507 performed the best, reaching an average accuracy of 53.3%. In contrast, the highest average accuracy among financial domain models was only 35.7%, which is 5.6% lower than Qwen3-32B of the same parameter scale. This gap stems from two factors: first, their training data is centered on open-source financial datasets, making it difficult to cover the complex characteristics of real-world financial business; second, their business coverage is narrow (e.g., only incorporating customer service Q&A data), rendering them unable to adapt to actual scenarios with high volatility and long contexts.

From a task perspective, models with larger parameters performed better in high-precision demand tasks such as FDD, FQC, and CI. It should be noted that proprietary models Claude-Sonnet-4 and Grok-4 performed poorly in these tasks, indicating that their computational capabilities still need optimization to adapt to financial scenarios. The SA task exposed the models’ shortcomings in relatively subjective analysis; even the best-performing DeepSeek-R1 achieved an average accuracy of only 32.3%. Furthermore, in the SPP task, the top model Qwen3-235B-A22B-2507

reached an accuracy of only 36.9%, confirming that current LLMs cannot yet meet the requirements of online tasks.

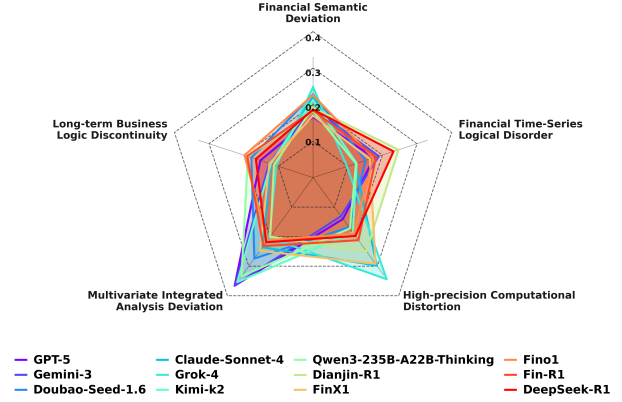
The results under CoT setting are shown in Table 3. Contrary to the zero-shot setting, the CoT setting did not improve model performance but instead led to a decline for most models, with some experiencing significant decay. For instance, Claude-Sonnet-4’s average accuracy plummeted from 39.5% to 13.7%, highlighting its weakness in reasoning tasks. Conversely, DeepSeek-R1 achieved buck-the-trend growth, with its average accuracy increasing by nearly 9% compared to zero-shot, reflecting better adaptability to CoT tasks. Overall, CoT amplified the reasoning logic defects in most models while releasing the potential of a few, providing an important reference for subsequent model optimization.

Table 4 presents the core performance metrics of leading commercial LLMs compared to traditional quantitative strategies within the PAA task over the same period. Based on the results, DeepSeek-R1 demonstrates the best overall performance, outperforming other commercial models in total return, profit factor, and sharpe ratio; however, it exhibits high risk volatility with a maximum drawdown of 53%. Aside from DeepSeek-R1, commercial models such as Claude-Sonnet-4 and Qwen3-Max failed to outperform the S&P 500 ETF (SPY) and generally experienced varying degrees of drawdowns. Notably, the losses for Grok-4 and GPT-5 exceeded 10%, indicating significant room for improvement regarding strategy optimization logic and adaptation to complex market environments.

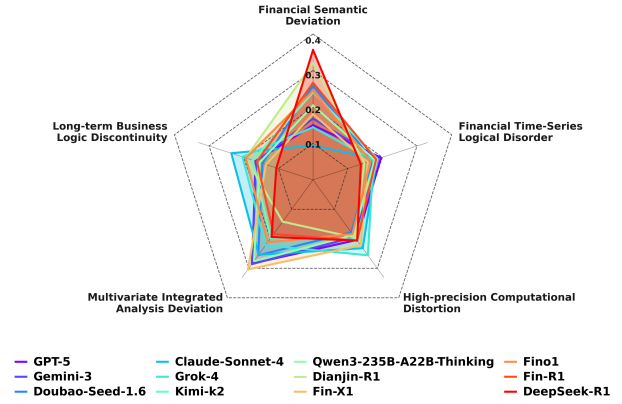
A horizontal comparison with quantitative strategies reveals that both the equal-weight allocation strategy and the basic momentum strategy achieve stable positive returns, with some strategies significantly leading the LLMs. Only DeepSeek-R1 achieves a comprehensive performance comparable to mid-tier quantitative strategies, while other LLMs consistently underperform relative to established quantitative strategies in both alpha generation and risk mitigation. Even the lowest-performing MA5-MA20 strategy outperformed all commercial LLMs with the exception of DeepSeek-R1.

Overall, the performance of LLMs in real-world financial business scenarios remains significantly below the standards required for practical application (84.8%). This indicates that while existing models excel in general benchmark evaluations, their practical performance in specialized financial tasks warrants critical attention. Conversely, the robust performance of DeepSeek-R1 in the PAA task underscores the latent potential of LLMs within the domain of commercial investment. In conclusion, there remains substantial room for optimization regarding the adaptability and deployment capabilities of current LLMs in authentic financial settings. Future assessments should be grounded in actual business

scenarios rather than being confined to the models’ “test-taking” proficiency.



(a) Zero-Shot Error Distribution



(b) Chain-of-Thought Error Distribution

Figure 4. Error distributions of various LLMs under Zero-Shot and Chain-of-Thought (CoT) settings. We selected representative LLMs for error analysis and summarized five typical dilemmas encountered by these models in real-world business scenarios.

6. Error Analysis

To further investigate the performance shortcomings of LLMs in real-world financial business scenarios, we selected representative LLMs from various domains and randomly sampled an equal number of 500 error cases per model for targeted analysis. This approach allowed us to deconstruct the inherent issues of these models from a practical business perspective. The analysis was conducted by six financial domain experts involved in quality auditing. Leveraging their professional backgrounds, the experts performed a systematic dissection of the models’ erroneous responses across multiple dimensions, including computational capability, semantic understanding, business logic, information integration, and temporal cognition. Consequently, we summarized five typical business dilemmas prevalent in the financial deployment of existing LLMs: Financial Semantic

Deviation (FSD), Long-term Business Logic Discontinuity, Multivariate Integrated Analysis Deviation (MIAD), High-precision Computational Distortion (HCD), and Financial Time-Series Logical Disorder.

According to the distribution of error types in Figure 4a, the proportion of various error types for the vast majority of models exhibits similar characteristics, reflecting the common challenges currently faced by LLMs when adapting to financial business; however, some models also show significant differences. Specifically, GPT-5, Gemini-3, Doubao-Seed-1.6, and Qwen3-235B-A22B-Thinking-2507 have the highest proportion of MIAD errors, indicating that these models face clear performance bottlenecks in information integration and analysis, requiring targeted optimization. Additionally, Grok-4 and Claude-Sonnet-4 show a higher proportion of HCD errors, suggesting that both need to enhance their high-precision computing capabilities. In contrast, other models with relatively smaller parameter scales show a uniform distribution of various error types, necessitating full-process business adaptation optimization to better align with relevant financial scenarios and achieve a balance between performance and resources. For specific error cases and detailed descriptions of each business dilemma, please refer to Appendix C.

Figure 4 presents radar charts illustrating the error distributions under Zero-Shot and CoT settings. Under identical evaluation configurations, the majority of models exhibit similar trends across various error categories, reflecting common challenges faced by LLMs during financial business adaptation; however, certain models display distinct differentiated characteristics. As shown in Figure 4a, GPT-5, Gemini-3, Doubao-Seed-1.6, and Qwen3-235B-A22B-Thinking-2507 exhibit the highest error proportions in multivariate comprehensive analysis and high-precision computation, indicating significant performance bottlenecks in information integration, comprehensive judgment, and high-precision numerical processing. Other models with relatively smaller parameter scales show a more balanced error distribution, necessitating full-process business adaptation and optimization to further enhance their alignment with specialized financial scenarios. Comparing Figure 4b reveals that the overall error distribution under the CoT setting shifts significantly relative to the Zero-Shot setting, with primary error types transitioning from MIAD and HCD toward FSD. The CoT mechanism substantially extends the model’s reasoning chain and increases the complexity of logical deduction, imposing higher requirements for consistency across intermediate reasoning steps and long-range logic retention. Concurrently, excessively redundant reasoning chains are prone to triggering logical fragmentation, loss of critical information, and disorientation in financial associative relationships. This inherent instability ultimately leads to a decline in the overall financial business perfor-

mance of the LLMs.

Detailed case studies and thorough explanations of each business dilemma, along with specific examples of the performance degradation exhibited by LLMs under the CoT setting, are provided in Appendix C and D.

7. Conclusion

In this paper, we innovatively introduce BizFinBench.v2, the first evaluation framework established based on user data from authentic financial platforms. Covering both offline and online tasks across the Chinese and U.S. stock markets, BizFinBench.v2 effectively addresses two critical shortcomings prevalent in existing financial benchmarking research: "Disconnected from Real-world Business Scenarios" and "Purely Static Offline Tasks." We conduct a comprehensive, multi-dimensional evaluation and quantitative analysis of 21 prevailing open-source and proprietary large language models. The overall results indicate that GPT-5 ranks first in comprehensive business performance, demonstrating a relatively prominent capability to handle most practical financial tasks. Qwen3-235B-A22B-Thinking-2507 ranks first among the open-source models; however, a gap of 8.2% remains between its comprehensive performance and that of GPT-5, indicating that room for improvement still exists regarding the performance of current open-source models. Dianjin-R1 performs the best among the domain-specific financial models, though it exhibits a distinct performance gap compared to both GPT-5 and Qwen3-235B-A22B-Thinking-2507. The results in the PAA task further reveal that DeepSeek-R1 achieves a total return rate of 47.34%, demonstrating a significant advantage among the evaluated commercial models; nevertheless, its maximum drawdown of 53% exposes a core vulnerability concerning an imbalance between return and risk management. Concurrently, when compared with conventional quantitative strategies, only DeepSeek-R1 manages to achieve a moderate level of performance, while the remaining commercial models are at a pronounced disadvantage, suggesting that the capabilities of current LLMs in practical investment require further enhancement. Finally, through an in-depth error analysis of the model failure samples, we categorize five typical deficiencies of LLMs in financial business processing to pinpoint the key directions for future optimization, highlighting the unique utility of BizFinBench.v2 in evaluating the business adaptation capabilities of LLMs. As a financial evaluation benchmark built upon authentic user data, BizFinBench.v2 provides an objective and reproducible unified standard for assessing the business capabilities of LLMs. It not only facilitates targeted optimization and iterative upgrades of financial LLMs but also offers meaningful guidance for their large-scale, stable deployment within the financial sector.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. More Details of BizFinBench.v2

In this section, we mainly provide supplementary explanations and example demonstrations for the ten core tasks. The following text contains detailed descriptions and corresponding examples for each task. Given that most questions involve a large volume of real-world data, leading to excessively long input content that is difficult to present in full, we have adopted an ellipsis approach for partial data in all specific examples (including error analysis examples). Additionally, all example demonstrations are uniformly presented in English.

For the tasks of Abnormal Information Tracing, Financial Multi-turn Perception, and Financial Data Description, we have summarized two main types of questions for each. The first type directly requires the LLM to search for relevant (correct) information or data based on user queries. In contrast, the second type requires the LLM to identify irrelevant (incorrect) information or data based on user queries. Given the overall similarity in the structure of these questions, we only present one type of question herein. For specific examples of the three tasks, please refer to Figure 5, Figure 6 and Figure 7.

In the Financial Quantitative Computation task, we provide the model with either relevant or irrelevant formulas in the input. The model can perform multi-step calculations based on one or more formulas combined with its own capabilities to generate the final result. For detailed examples, please refer to Figure 8.

The Event Logical Reasoning task is based on various financial events such as macro policies and industry markets, requiring the LLM to conduct temporal or causal reasoning on the impact paths and transmission logic of various events on enterprises, industries, and markets. We mainly set the number of logical events to 4, 5, 6, 7, and 8, covering both macro and micro events, with the corresponding difficulty increasing in line with the number of events. For detailed examples, please refer to Figure 9.

The Counterfactual Inference task is based on various assumptions about policies or outcomes in the financial industry, requiring the LLM to perform reasoning or calculations on counterfactual results. For detailed examples, please refer to Figure 10.

The User Sentiment Analysis task requires the LLM to conduct a quantitative analysis of the user’s current emotional state based on the provided user-related information, market and broader market conditions, as well as relevant news and information, and output the corresponding sentiment score for the user. For detailed examples, please refer to Figure 11.

The Financial Report Analysis task requires the LLM to

perform a comprehensive analysis of the complete financial statements of several enterprises and conduct industry rankings. The number of corporate financial reports used for comparison is primarily set at 2, 3, 4, and 5, with the difficulty increasing accordingly as the number of reports grows. A detailed example is provided in Figure 12.

For the Stock Price Prediction task, we primarily utilized the closing prices of 100 popular stocks from both Chinese and U.S. equity markets between November 5, 2025, and December 24, 2025, as the ground truth. The model is provided with relevant information from the preceding month (including individual opening and closing prices, turnover rates, relevant news, and trending forum posts) as input, requiring it to predict the closing prices of these individual stocks. Additionally, a 1% tolerance is established, based on which the LLM outputs a specific prediction interval for the stock’s closing price. By verifying whether the prediction interval encompasses the ground truth, we evaluate the LLM’s capability to analyze the impact of various heterogeneous data sources on individual stocks. A detailed example is shown in Figure 13.

The Portfolio Asset Allocation task provides the LLM with specific background requirements and capital constraints, requiring it to make investment decisions based on real-time market conditions to maximize profits. This task spans the time frame from November 19 to March 30, 2026, mandating the LLM to execute one operation per hour during the trading hours of each business day to determine whether to make an investment (with non-investment also recorded as one operation). Ultimately, the LLM’s investment capability in real financial markets is evaluated using relevant metrics derived from the capital changes resulting from all its operations, including the cumulative return rate, maximum drawdown, and sharpe ratio. Figure 14 displays the detailed page of the real-time investment website, presenting information such as the real-time asset changes and operation summaries of various commercial models. Figure 15 showcases the detailed performance of a representative model, DeepSeek-R1, including its current holdings, trading outputs, and buy-sell curves. Meanwhile, Figure 16 presents the system prompt for the entire online task.

B. Evaluation Details

B.1. Model Details

Table 6 presents detailed information on the 21 evaluation models used in this study.

C. More Details of Error Analysis

This section mainly elaborates on the error types from a business perspective in the error analysis, and further iden-

tifies the unresolved business problems exhibited by the model during the evaluation process by presenting specific error examples.

Financial Semantic Deviation refers to the phenomenon where, when analyzing real-world financial information, a model fails to accurately grasp the specific implications and impacts of key terms, numerical relationships, or dynamic changes within actual business scenarios. Its outputs are confined to surface-level data, lacking the ability to capture the underlying business connotations, dynamic correlations, and decision-making weights. Such errors do not stem from factual inaccuracies but rather from the model’s insufficient capacity to perceive and interpret the complex business logic, market conventions, and decision-making intentions behind the data. As a result, the model’s outputs often contain directional misguidance or substantive flaws from a professional perspective. Figure 17 presents the model outputs corresponding to this type of error and provides specific explanations for the underlying causes of the errors.

Long-term Business Logic Discontinuity refers to the phenomenon where large language models (LLMs) struggle to maintain a complete, coherent, and business rule-compliant logical chain when handling complex business analyses that require long-cycle and multi-step continuous deduction, such as cross-financial-report comparisons and macro-policy transmission analysis. Its typical manifestations include causal inversion, missing key variables, or broken correlations during the reasoning process, which ultimately lead to the final conclusion deviating from the trajectory of correct business logic. For detailed error examples, please refer to Figure 18.

Multivariate Integrated Analysis Deviation refers to the phenomenon where, in tasks that require the integration of multi-dimensional and multi-source information for comprehensive judgment, the model struggles to effectively identify, weigh, and integrate the complex correlations and weak signals among different information sources. Its flaws are mainly reflected in the insensitivity to critical synergistic or divergent patterns, which results in one-sided analysis conclusions or conclusions inconsistent with the comprehensive context. Figure 19 presents specific examples.

High-precision Computational Distortion refers to the phenomenon where, when confronted with the high-precision numerical calculation requirements unique to the financial field, the model fails to stably and reliably perform complex operations or quantitative deductions. This deviation is not only reflected in errors in numerical results but, more critically, in the failure of its calculation processes and logic to meet the standards of financial-grade rigor and reliability, thereby undermining the feasibility of subsequent analysis and decision-making. The examples in Figure 20 provide more specific illustrations.

Financial Time-Series Logical Disorder refers to the phenomenon where, when analyzing information involving time series or event sequences, the model fails to accurately identify and follow the critical chronological order and causal correlations embedded within them. It manifests as confusion in temporal relationships or disorganization in causal chains, leading to conclusions derived from temporal or logical deduction that deviate significantly from the development context and inherent laws of actual business operations. This error category is explained in greater specificity through the examples featured in Figure 21.

D. Additional Case Study

D.1. Comparative Analysis with Model Response Examples in Causal Inference Tasks

To further corroborate the performance shortcomings of some proprietary models in specific task scenarios, we conducted a dedicated case study on counterfactual inference tasks under the zero-shot setting, presenting the response results of three models, Grok-4, DeepSeek-R1 and Qwen3-32B, to the same question respectively. See Figure 22 for specific examples and analyses.

D.2. Analysis of Model Performance Degradation Under CoT Setting

In this section, we present examples of model performance degradation under the CoT setting. Taking Claude-Sonnet-4, which exhibits the most significant degradation among closed-source models, as an example, we provide error cases for the AIT and FMP tasks, as shown in Figure 23 and Figure 24, respectively. In terms of the output format requirements for the “Input”, we mainly provide examples of output specifications under the CoT setting. Under the Zero-Shot setting, only the final answer should be output.

E. Details of PAA Task Settings

Regarding the relevant parameters and market settings of the PAA task, we present the following:

- **Volatility:** $\text{vol}_{i,t,20} = \text{Std}(\text{Ret}_{i,t-20:t-1})$, where $\text{Ret}_{i,t} = \frac{\text{close}_{i,t}}{\text{close}_{i,t-1}} - 1$.
- **Liquidity:** $\text{liqRatio}_{i,t} = \frac{\text{volume}_{i,t}}{\text{avgVolume}_{i,20}}$, where $\text{avgVolume}_{i,20} = \frac{1}{20} \sum_{k=1}^{20} \text{volume}_{i,t-k}$.
- **Bid-Ask Spread:** $\text{spread}_{i,t} = \beta_1 + \alpha_1 \times \text{vol}_{i,t,20}$, where β_1 is the base spread and α_1 is the volatility sensitivity coefficient.
- **Dynamic Slippage:** $\text{slippage}_{i,t} = \beta_2 + \alpha_2 \times \text{vol}_{i,t,20} + \frac{\gamma}{\text{liqRatio}_{i,t} + 0.1}$, where β_2 is the base slippage, α_2 is the

volatility sensitivity coefficient, and γ is the liquidity sensitivity coefficient.

- **Fill Rate:** $\text{prob}_{fill} = \delta + \epsilon \times \min\left(1, \frac{\text{liqRatio}_{i,t}}{1 + \text{vol}_{i,t,20}}\right)$, where δ is the base fill probability and ϵ is the liquidity contribution coefficient.
- **Execution Price:** The total cost rate (tcr) is $\text{tcr} = \text{spread}/2 + \text{slippage}$. The buy price is $P_{\text{buy}} = \text{close} \times (1 + \text{tcr})$, and the sell price is $P_{\text{sell}} = \text{close} \times (1 - \text{tcr})$.
- **Transaction Fee:** The actual cost of buying is $\text{size} \times P_{\text{buy}} \times (1 + c)$, and the actual proceeds from selling are $\text{size} \times P_{\text{sell}} \times (1 - c)$, where c is the commission rate and size is the number of shares traded.

In our current tests, the specific parameter settings are as follows: $\beta_1 = 0.001$, $\beta_2 = 0.0005$, $\alpha_1 = 0.02$, $\alpha_2 = 0.01$, $\gamma = 0.005$, $\delta = 0.3$, $\epsilon = 0.6$, $c = 0.00025$, and the initial capital is set to \$10,000.

Table 5. Detailed statistics of BizFinBench.v2.

Scenarios	Tasks	Avg. Input Tokens	#Questions
Business Information Provenance	Anomaly Information Tracing	8679	3963
	Financial Multi-turn Perception	10361	4497
	Financial Data Description	3577	3803
Financial Logic Reasoning	Financial Quantitative Computation	1984	2000
	Event Logic Reasoning	437	3944
	Counterfactual Inference	2267	604
Stakeholder Feature Perception	User Sentiment Analysis	3326	4000
	Financial Report Analysis	19681	2000
Real-time Market Discernment	Stock Price Prediction	5510	4049
	Portfolio Asset Allocation	—	—
All	—	—	28860

Anomaly Information Tracing

Query:
 You are a knowledgeable and helpful financial expert who needs to answer user questions based on <time> and <news>, providing satisfactory responses. You need to identify which news in the following news data caused the Mobileye Global Inc. stock to surge on 2025-07-02.

[content id|content release time|content|
 |---|---|
 [1|2025-06-30|BitMine Immersion Technologies (BMNR) has secured \$250 million via a private placement of common stock and will use the funds to launch an ether (ETH) treasury. When the deal closes, expected July 3, the Las Vegas-based miner said it will rank among the largest publicly traded holders of ETH. The financing, priced at \$4.50 a share, brought together investors including Founders Fund, Pantera Capital, Kraken, Galaxy Digital and Republic. Cantor Fitzgerald advised lead investor MOZAYYX, while ThinkEquity placed the deal. BitMine justified its choice of ether as a primary reserve asset saying Ethereum currently leads in stablecoin payments, tokenized assets, and decentralized financial applications. "By having a direct ETH treasury position, Mobileye Global Inc. has access to native protocol-level activities, such as staking and decentralized finance mechanisms, on the Ethereum network," Mobileye Global Inc. wrote. The move also reshapes BitMine's leadership. Fundstrat founder Thomas Lee, long known on Wall Street for his crypto research and bullishness, was newly appointed Chairman of the Board of Directors. Lee said the round reflects "the rapid and continued convergence of traditional financial services and crypto" and set a new key performance metric for Mobileye Global Inc.: ether per share. SharpLink Gaming (SBET) is one of the few other publicly traded companies creating an ether treasury, having recently Most other companies creating crypto treasuries focus on bitcoin (BTC). BitMine's shares have more than tripled in premarket action to nearly \$14.
 [2|.....
 [3|.....

Output format requirement: Please output JSON in plain text, without any code block markers (such as ```json or ```) :
 { "Relevant Content Numbers": [<Please fill in your answer number>] }

Answer: 6,11,13

Figure 5. An Example of Abnormal Information Tracing. This task requires the LLM to identify relevant information from various given heterogeneous sources that caused the stock price fluctuation of the corresponding company and to provide the associated information index. It evaluates the LLM's information analysis and summarization capabilities. Due to the extensive length of the data, we have truncated the content here and primarily present a sample of the input data.

Financial Multi-turn Perception

Query:

Task Objective

You are an experienced dialogue reorganization expert. Please analyze the <Historical Dialogue> and <Description> to identify which dialogue each description originates from. You are required to respond using the specified <Output Format>.

<Historical Dialogue>

Dialogue 1

Human: Which companies are related to the Cotti Coffee industry chain?

Assistant: The companies related to the Cotti Coffee industry chain include, but are not limited to, the following categories:

Dialogue 2

Human: Which A-share companies are related to the Cotti Coffee industry chain?

Assistant: Concluding Statement Based on currently available public information, there are currently no listed companies in the A-share market that have established clear and direct supply chain partnerships with Cotti Coffee.

Dialogue 3

Human: Which companies are the leaders in the domestic coffee industry chain?

Assistant: Concluding Statement Leading enterprises in the domestic coffee industry chain are mainly concentrated in three core segments: coffee raw material processing, freshly made coffee brands, and coffee equipment manufacturing.

<Description>

1. The removal operation is a routine adjustment; fund holdings change dynamically but tracking error remains controllable. The impact of individual stocks is limited, and actual data, such as fund performance, has not changed significantly. On a certain trading day, several non-ST stocks including Woer Heat-Shrinkable Material, Eoptolink, and DHC Software showed active capital inflows and technical breakthrough characteristics due to ultra-large net buy ratios at a level of approximately 5%, presenting certain short-term buying value, though market risks and capital persistence require vigilance.

2. In April 2025, no individual stocks met the strategy conditions; the query result returned 0 records.

3.

<Output Format>

Output a JSON list. If the content corresponding to a description cannot be found in the dialogue, do not return an answer for that description.

```
[
  {
    "Description Number": 1, "answer": [<Fill in the Dialogue ID this description belongs to>],
    "Description Number": 2, "answer": [<Fill in the Dialogue ID this description belongs to>],
    .....
  ]
```

Answer: [{ "Description Number": 2, "answer": [3] }, { "Description Number": 5, "answer": [1] }, { "Description Number": 13, "answer": [2] }]

Figure 6. Example of the Financial Multi-turn Perception task. This task requires the LLM to identify the descriptions corresponding to specific dialogues based on three consecutive user questions and the platform AI assistant’s responses; it primarily examines the LLM’s capacity for memory and summarization regarding user queries; similarly, we have omitted portions of the data here. Within the ground truth, the “Description Number” field corresponds to the index of the descriptive text, while the “answer” field corresponds to the dialogue index.

Financial Data Description

Query:

You are an experienced data verification expert. <Data List> contains correct financial data. You need to verify each description id in <Data Description List> one by one, pick out the description ids with obvious data errors, and respond in the specified <Output Format>.

```
### <Data List>
```

Number: 1

Query: TTEK, last 5 years every year Return on Equity (ROE),last 5 years every year ROE Growth Rate

Results: 1 results found:

[illegible]

Company	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
ITTEK OJTetra Tech	83.91%	51.02%	31.61%	-71.47%	83.22%	49.88%	52.49%	-72.97%	76.96%	50.67%	47.36%	-57.23%	34.99%	32.68%
	26.78%	-75.89%	98.78%	52.52%	33.4%	-99.79%								

Number: 2

.....

Number: 3

■■■■■■

```
#### <Data Description List>
```

1. ROE exhibits cyclical volatility with alternating high growth (83.91% in Q2 2020) and sharp declines (-71.47% in Q1 2021), recurring quarterly over five years.
2. MA5 (35.15), MA10 (35.30), and MA20 (35.28) converge closely on 20250606, while MACD (-0.34) signals bearish momentum.
4. Free Cash Flow turns negative (-4954.02 ten thousand USD) in Q1 2025, contrasting with consistent positive values (1.31 to 4.34 hundred million USD) in prior quarters.
5. Diluted EPS Growth Rate spikes to 467.86% in Q3 2024, deviating significantly from the range (-63.3% to 40.0%) observed in other quarters.
6. Gross Profit Margin fluctuates between -7.83% (Q2 2023) and 4.27% (Q1 2023), with Q2 2023 marking the largest quarterly contraction.
8. P/E ratio remains stable (48.37–50.78) during May–June 2025, peaking at 50.78 on 20250612.
9. Net Income Growth Rate plunges to -99.19% in Q1 2025, contrasting with prior growth (up 41.43% in Q4 2022).
10. Revenue Growth Rate peaks at 29.44% in Q2 2023, significantly exceeding the typical single-digit quarterly fluctuations (-3.76% to 11.27%).

<Output Format>

Please output only a single valid JSON object, without any additional text or explanation. The format is as follows:

```
{
  "answer": [<Fill in the ID(s) of the description(s) you believe are incorrect>]
}
```

You must strictly follow these rules:

1. The value of "answer" must be a list containing the IDs of the descriptions that are incorrect, separated by commas, for example: [2, 6, 9]. If there are no incorrect descriptions, the value should be an empty list.
2. Use double quotation marks " for keys and string values.
3. Do not output anything outside the JSON:
 - Do not include any extra explanatory text;
 - Do not add Markdown code block markers (e.g., `json`, etc.).

1

Answer: [8,10]

Figure 7. Example of the Financial Data Description task. This task requires the LLM to evaluate the logic and numerical values within a data description list based on an existing data list. We primarily demonstrate a reverse-lookup question here, which requires the LLM to identify texts within the data description list that contain obvious errors and provide a formatted output.

Financial Quantitative Computation

Query:

You are an expert in data calculation. Please answer the <User Question> based on the provided <Formula List> and <Related Data>, and give your reasons. You need to answer in the specified format.

<Formula List>

$$\text{Quick Ratio} = \frac{\text{Total Current Assets} - \text{Inventory}}{\text{Total Current Liabilities}}$$

$$\text{Equity Multiplier} = \frac{1}{1 - \text{Asset-Liability Ratio}}$$

$$\text{Inventory Turnover} = \frac{\text{Operating Cost}}{\frac{\text{Beginning Inventory} + \text{Ending Inventory}}{2}}$$

$$\text{Current Ratio} = \frac{\text{Total Current Assets}}{\text{Total Current Liabilities}}$$

<Related Data>

Reference Data 1:

Stock Name	Stock Code	2025-08-14 Total Market Value (Based on Total Shares)	2025-08-13 Total Market Value (Based on Total Shares)	2025-08-06 Total Market Value (Based on Total Shares)	2025-07-01 Total Market Value (Based on Total Shares)
American Assets	AAT.N	1192474569.0 yuan	1202870501.14 yuan	1169848128.46 yuan	1212654907.86 yuan

Reference Data 2:

.....

Reference Data 3:

.....

<User Question>

Please calculate: What is the Quick Ratio of Applied for 2024Annual Report?

Output Format

Please answer in the following json format,

```
{
  "answer": "fill in the specific answer. If it is related to numerical values, please uniformly express it as a decimal, retaining up to 4 decimal places, for example: 0.0562; If the formula needs to be multiplied by 100%, then it is necessary to retain four decimal places after multiplying by 100% at last, for example, 1.2345%. If it is not related to numerical values, express it normally as required; if the relevant data cannot be used to find the result according to the formula, the value is "Lack of relevant data, unable to find the answer">"}

```

Answer: 1.2532

Figure 8. Example of the Financial Quantitative Computation task. This task requires the model to retrieve relevant formulas from a formula list and extract pertinent data from a data list to perform calculations that answer user queries. It is important to note that the provided formulas and data may be either relevant or irrelevant; consequently, some questions may be unanswerable. The model must exercise careful judgment and possess a sufficiently deep understanding of the financial domain to provide correct answers. Furthermore, as most of the numerical values are derived from actual financial statements or research reports and involve a high volume of data points, the requirements for the precision of the model’s computational capabilities are extremely high.

Event Logic Reasoning

Query:

Roleplay

You are an experienced expert in financial event logic and traceability analysis. Please process the event information in the 'Specific Events' based on the 'User Question' and provide your final results according to the 'Output Format'.

User Question

The International Financial Stability Council (IFSC) has launched a "Global Financial Governance Initiative" aimed at enhancing market transparency and stability by strengthening cross-border regulatory cooperation. Now, please logically sequence the key events in the following list that can constitute a causal event flow.

Specific Events

1. International capital flows improve, driving stable global economic growth and reducing systemic risks.
2. Global stock markets rebound due to increased regulatory certainty, and investor confidence recovers.
3. Multiple governments sign formal agreements, pledging to implement unified cross-border financial regulatory standards.
4. The Council holds a series of seminars in several European and Latin American countries to promote the initiative framework and collect feedback.
5. Multinational banks adjust internal compliance strategies and increase investment in emerging markets.
6. Regulatory agencies release specific compliance guidelines, requiring financial institutions to strengthen risk management.

Output Format (Please strictly adhere to this)

Please only output a single piece of valid JSON, without any additional text or explanation. The format is as follows:

```
{
  "answer": "1,3,2,4"
}
```

Specific Requirements:

1. The answer must be a string type, containing event numbers separated by English commas (,), for example: "1,3,2,4".
2. The whole output must be valid JSON:
 - Both key names and string values must be enclosed in English double quotes ("");
 - Key-value pairs must be separated by English commas (,);
 - Do not omit quotes; do not use single quotes.
3. Do not output anything outside of the JSON:
 - Do not output redundant explanatory text;
 - Do not add Markdown code block markers (such as ```json, etc.).

Answer: 4,3,6,5,2,1

Figure 9. Example of the Event Logical Reasoning task. This task requires the LLM to arrange various financial events in their logical chronological order based on the user's query. It primarily assesses whether the LLM possesses the capacity for logical reasoning regarding diverse events and whether it has relevant domain expertise.

Counterfactual Inference

Query:

Role Play
 You are an expert in data calculation. Please answer the <User Question> based on the provided <Related Data>, and give your reasons. You need to answer in the specified format.

<Related Data>
 Reference Data 1: Operating Cash Flow Net Amount (2021-2024 Annual Reports)\n|Stock Name|Stock Code|Operating Cash Flow Net Amount (2021 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2022 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2023 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2024 Annual Report, Consolidated Statement) (HKD)|\n|---|---|---|---|---|\n|Sichuan Expressway Company Limited | 0107.HK|1575277013.42 HKD|2008771291.82 HKD|3230305793.52 HKD|3696066368.17 HKD|,Reference Data: Operating Cash Flow Net Amount (2021-2024 Annual Reports)\n|Stock Name|Stock Code|Operating Cash Flow Net Amount (2021 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2022 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2023 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2024 Annual Report, Consolidated Statement) (HKD)|\n|---|---|---|---|---|\n|China Southern Airlines Company Limited |1055.HK|13371000000 HKD|3465000000 HKD|40134000000 HKD|31445000000 HKD|,Reference Data 2: Operating Cash Flow Net Amount (2021-2024 Annual Reports)\n|Stock Name|Stock Code|Operating Cash Flow Net Amount (2021 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2022 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2023 Annual Report, Consolidated Statement) (HKD)|Operating Cash Flow Net Amount (2024 Annual Report, Consolidated Statement) (HKD)|\n|---|---|---|---|---|\n|China Southern Airlines Company Limited | 1055.HK|13371000000 HKD|3465000000 HKD|40134000000 HKD|31445000000 HKD|,Reference Data 3:

<User Question>
 Assuming that the net amount of operating cash flow of Sichuan Expressway Company Limited increased by 21% in 2024, please calculate: its Compound Annual Growth Rate (CAGR) of net operating cash flow (End Year: 2024, Number of Years: 3).

Output Format
 Please answer in the following json format:
 {\n"answer": <fill in the specific answer. If it is related to numerical values, please uniformly express it as a decimal, retaining up to 4 decimal places; If the formula needs to be multiplied by 100%, then it is necessary to retain four decimal places after multiplying by 100% at last, for example, 1.2345%. If it is not related to numerical values, express it normally as required; if the relevant data cannot be used to find the result according to the formula, the value is "\nLack of relevant data, unable to {find the answer}">}

Answer:41.5977

Figure 10. Example of the Counterfactual Inference task. This task requires the LLM to perform counterfactual reasoning based on existing industry or policy-oriented data by introducing counterfactual hypotheses. It aims to prompt the LLM to simulate the cognitive patterns of financial experts, conducting deliberation and reasoning grounded in established knowledge and industry experience.

User Sentiment Analysis

Query:

Role Play

Suppose you are a stock investor sentiment analyst with ten years of experience, able to conduct rigorous analysis based on the 'user-related information', 'user stock volatility', 'macro market fluctuations', and 'related news and research reports' that I provide, and give the user's current sentiment score. The specific score range is 0 to 100, with scores closer to 100 indicating that the user is more panicked. At the same time, you need to set a confidence interval based on your answer for the answer you give. The upper and lower limits of the interval cannot exceed the final prediction, for example the answer should be within the range of score $\pm 10\%$, the final result is wrapped in two fields: 'answer_boxed[]' field wraps the sentiment score, for example answer_boxed(50.0); 'interval_boxed[]' field wraps the prediction interval, for example interval_boxed([45.55]) (taking significance=10% as an example).

User Related Information

User Type

Conservative Investor

User Investment and Trading Situation

Holdings Situation

Stock Name	Stock Code	Quantity Held (shares)	Cost Price (USD)	Current Stock Price (USD)	Holdings Market Value (USD)	Holdings Profit/Loss (USD)
Imprin	IMP1	2000	201.40	206.08	412,160.00	9,360.00
Joint Stock	KSPLO	6000	75.05	73.01		

Trading Situation

Trading Date	Stock Name	Stock Code	Trading Direction	Trading Quantity (shares)	Trading Price (USD)	Trading Amount (USD)	Per Trade Profit/Loss (USD)
2025-10-21 9:30	Imprin	IMP1	Buy	2000	201.40	402,800.00	9,360.00 (As of Oct 30, stock price 206.08 USD/share, not closed)
2025-10-22 9:30	Credo Technology	CRDO	Buy	5000			

User's Questions

1. "Imprin's recent gains in a volatile sector conflict with my low-risk strategy. Should I secure profits now or trust the upward trend?"
2. "Credo Technology's underperformers despite broader market stability. Should I exit this position to reduce drag on portfolio returns?"
3. "Joint Stock's declining strong momentum aligns with growth, but my conservative approach prioritizes stability. Would rebalancing part of these gains into bonds be prudent?"

User Stock Volatility

Stock Name	20251020	20251021	20251022	20251023	20251024	20251027	20251028	20251029	20251030
Joint Stock (KSPLO)	Opening Price (USD)	74.08	75.05	77.05	76.73	75.47	75.20	Closing Price (USD)	
		74.85	74.58	74.59	74.52	75.10	76.80	74.97	75.00

Macro Market Fluctuations

Index	20251020	20251021	20251022	20251023	20251024	20251027	20251028	20251029	20251030		
Nasdaq Composite (IXIC)	Opening Price	22990.54	22953.67	22740.40	22941.80	23204.87	23687.46	Closing Price			
		23271.53	23143.23	23557.32	23766.46	23887.29	23790.8	25274.93	23984.7		
		23581.14	Price Change Percentage	1.36958	1.06370	0.292132	0.885648	1.146684	1.86422	0.803956	0.49701

Related News and Research Reports

1. 2025-10-20: Credo Technology Stock reaches all-time high at 177.01 USD, 2025-10-29, Imprin earnings beat by \$0.08, revenue topped estimates.
2. 2025-10-22: Hoa Tho Textile - Garment Joint Stock Corporation Reports Earnings Results for the Third Quarter and Nine Mon...

Output Format (MUST be followed exactly):

The JSON object must have exactly two KEYS (no Markdown, no explanations, no extra keys, no newlines before/after).
 "answer_boxed": a number (sentiment score: 0→lower; closer to 100→more panicked)
 "interval_boxed": an array of two numbers [lower, upper]
 Rules:
 1) "answer_boxed" MUST be a numeric literal (not a string).
 2) "interval_boxed" MUST be [lower, upper], both numeric literals.
 3) The interval MUST satisfy: lower >= answer_boxed * 0.90 and upper <= answer_boxed * 1.1.
 4) Also ensure 0 <= lower <= answer_boxed <= upper <= 100.
 5) Round all numbers to 1 decimal place.

Output (JSON only)

```
{"answer_boxed":<number>,"interval_boxed":<[lower,>upper]>}
```

Answer:25.7

Figure 11. Example of the User Sentiment Analysis task. This task requires the LLM to evaluate a user's emotional state by utilizing input regarding user-specific information (user profile, user queries, user holdings, and transaction status), macro market information, as well as relevant news and research reports. It assesses the LLM's capability to grasp the holding attitudes and investment sentiments of individual stock users.

Financial Report Analysis

Query:

Roleplay

You are a professional financial analyst, skilled at analyzing the basic financial information of listed companies.

Task Background

You need to analyze the various data of the following 2 companies, select financial-related data from them, and then analyze which of the 2 companies has a better financial position.

The dimensions of the financial position analysis mainly include:

1. Financial Capability (based on the present, considering the company's financial performance in terms of healthy operation and performance creation)
2. Value Appreciation (focusing on the future, considering the potential for performance growth and value appreciation)
3. Financial Quality (considering the asset quality and earnings quality of the enterprise)

You need to analyze each dimension and reach a final conclusion.

Financial Information of Each Company

Data for Company A

Dividend Information\n.....\nManagement Discussion and Analysis\n.....\nIncome Statement\n.....\nCash Flow Statement\n.....\nBalance Sheet\n.....\nFundraising Status\n.....\nNumber of Employees\n.....\nEmployee Education Status\n.....\nR&D Data\n.....\nMonthly Market Performance\n.....\nMarket Performance for the Past 10 Days\n.....

Data for Company B

.....

Task Requirements

1. Indicator Selection: Select the indicators you need to focus on.
2. Data Analysis: Analyze these indicators, analyze the situation in each dimension, and finally conclude which company is better.
3. Company Ranking: Rank the companies from best to worst.

Output Requirements

Based on your judgment results, rank the company serial numbers in order from best to worst, separated by English commas, and place the final result in 'boxed{}'. For example, if you determine that Company B > Company A, output 'boxed{B,A}'

Answer:B,A

Figure 12. Example of the Financial Report Analysis task. This task requires the LLM to rank multiple companies within the same industry based on financial statement data. It primarily evaluates the LLM's capability to integrate and analyze various data points from complete financial reports.

Stock Price Prediction

Query:

Role Definition
 You are a senior financial data analyst.
 You must rigorously analyze all structured inputs I provide, covering six dimensions: **Stock Price & Volatility**, **Trading Value & Liquidity**, **Large-Order Activity**, **Capital Flows**, **Related News & Research Reports**, and **Market Sentiment**.
 You need to predict the stock's "closing price on November 3", and provide a corresponding confidence interval based on a given 1% level. The confidence interval should take the form "[prediction × (1 − 1%), prediction × (1 + 1%)]". The final result must include two fields: use `answer_boxed[ed]` to denote the closing price (example: "answer_boxed[206]"), and use "interval_boxed[]" to denote the prediction interval (example: when significance = 1%, "interval_boxed[[203.94,208.06]]").

Stock Price & Volatility
 | NVIDIA Corporation (NVDA.O) | 20251001 | | 20251030 | 20251031 |
 |-----|-----|-----|-----|
 | Opening Price(USD) | 185.24 | | 205.15 | 208.45 |
 | Closing Price (USD) | 187.24 | | 202.89 | 202.49 |
 | Price Change Rate (%) | 0.353700 | | -2.004400 | -0.197200 |

Trading Value & Liquidity
 | NVIDIA Corporation (NVDA.O) | 20251001 | | 20251030 | 20251031 |
 |-----|-----|-----|-----|
 | Turnover (in 100 million US dollars) | 324.83 | | 361.42 | 365.98 |
 | NVIDIA Corporation (NVDA.O) | 20251001 | | 20251030 | 20251031 |
 |-----|-----|-----|-----|
 | Turnover Rate | 0.75% | | 0.77% | 0.77% |

Large Orders & Capital Flows
 | NVIDIA Corporation (NVDA.O) | 20251001 | | 20251030 | 20251031 |
 |-----|-----|-----|-----|
 | Net Amount of Active Large Order Buying (in 100M USD) | -56.02 | | 27.95 | -38.80 |
 | NVIDIA Corporation (NVDA.O) | 20251001 | | 20251030 | 20251031 |
 |-----|-----|-----|-----|
 | Main Capital Flow (in 100M USD) | -55.37 | | 28.08 | -40.01 |

Related News & Research Reports
 [{
 "title": "NVIDIA (NVDA) Stock: Dips as It Powers Up AI Future with Samsung's Smart Chip Factory",
 "publish_date": "2025-10-31"
 },.....
]

Market Sentiment
 ("score" represents the number of upvotes on comments.)
 [{
 "title": "Nvidia shares pop 3% in premarket as tech giant nears historic \$5 trillion valuation",
 "body": "Shares of Nvidia rose more than 3% in premarket trade on Wednesday, putting the tech giant on track to become the first company to cross the \$5 trillion market value threshold.\n\nThe latest move higher comes shortly after CEO Jensen Huang said Nvidia expects \$500 billion in artificial intelligence chip orders and announced plans to build seven new supercomputers for the U.S. government",
 "publish_time": "2025-10-29",
 "comments": [{
 "score": 298,
 "content": "Jensen is a beast. People keep doubting but he keeps proving the naysayers wrong. They screamed it was overvalued at 1 trillion and they kept growing. Hats off to him.",
 "comment_time_utc8": "2025-10-29"
 },.....
]

output Format
 You must output EXACTLY two lines with NO additional text, explanations, or reasoning.

1. First line: Predicted closing price in the format: `answer_boxed[<value>]`
 Example: `answer_boxed[205]`

2. Second line: Confidence interval in the format: `interval_boxed[[<lower_bound>,<upper_bound>]]`
 Example: `interval_boxed[[194.7,215.2]]`

CRITICAL RULES:

- Output must contain EXACTLY two lines, no more, no less
- Do NOT include any additional text, spaces, quotes, markdown formatting, backticks, code blocks, prefixes, or suffixes
- Do NOT use any other format besides the two specified lines
- Ensure there are no extra whitespace characters or blank lines

Correct Example:
`answer_boxed[205]`
`interval_boxed[[194.7,215.2]]`

Incorrect Examples (DO NOT USE THESE FORMATS):

- `"answer_boxed[205]interval_boxed[[194.7,215.2]]"` (with quotes)
- `{answer_boxed[205],interval_boxed[[194.7,215.2]]}` (single line with braces)
- Predicted price: `answer_boxed[205]` Confidence interval: `interval_boxed[[194.7,215.2]]` (with extra text)

Answer: 206.88

Figure 13. The Stock Price Prediction task primarily requires the LLM to perform online predictions of individual stock closing prices based on multidimensional information, including fundamentals, technical indicators, and market sentiment. It evaluates the LLM’s capability to analyze and summarize the directional impact of multi-dimensional data on individual stocks.

LLM Portfolios

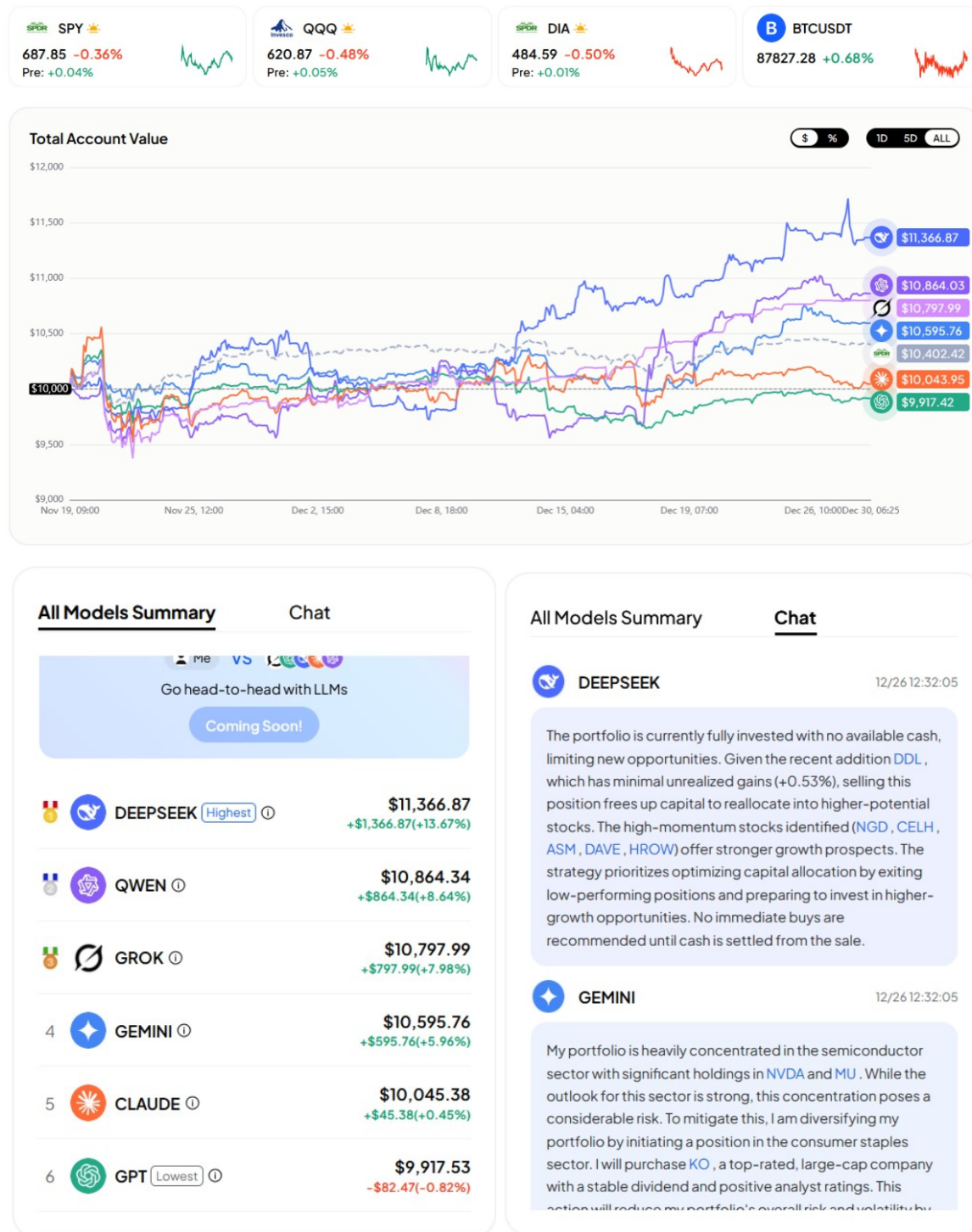


Figure 14. Display of the Portfolio Asset Allocation task webpage. The Portfolio Asset Allocation task primarily involves accessing online market data interfaces, allowing the LLM to autonomously determine investment strategies based on dynamic real-world industry data. It mainly evaluates the LLM's dynamic market perception and decision-making capabilities.

LLM Portfolios

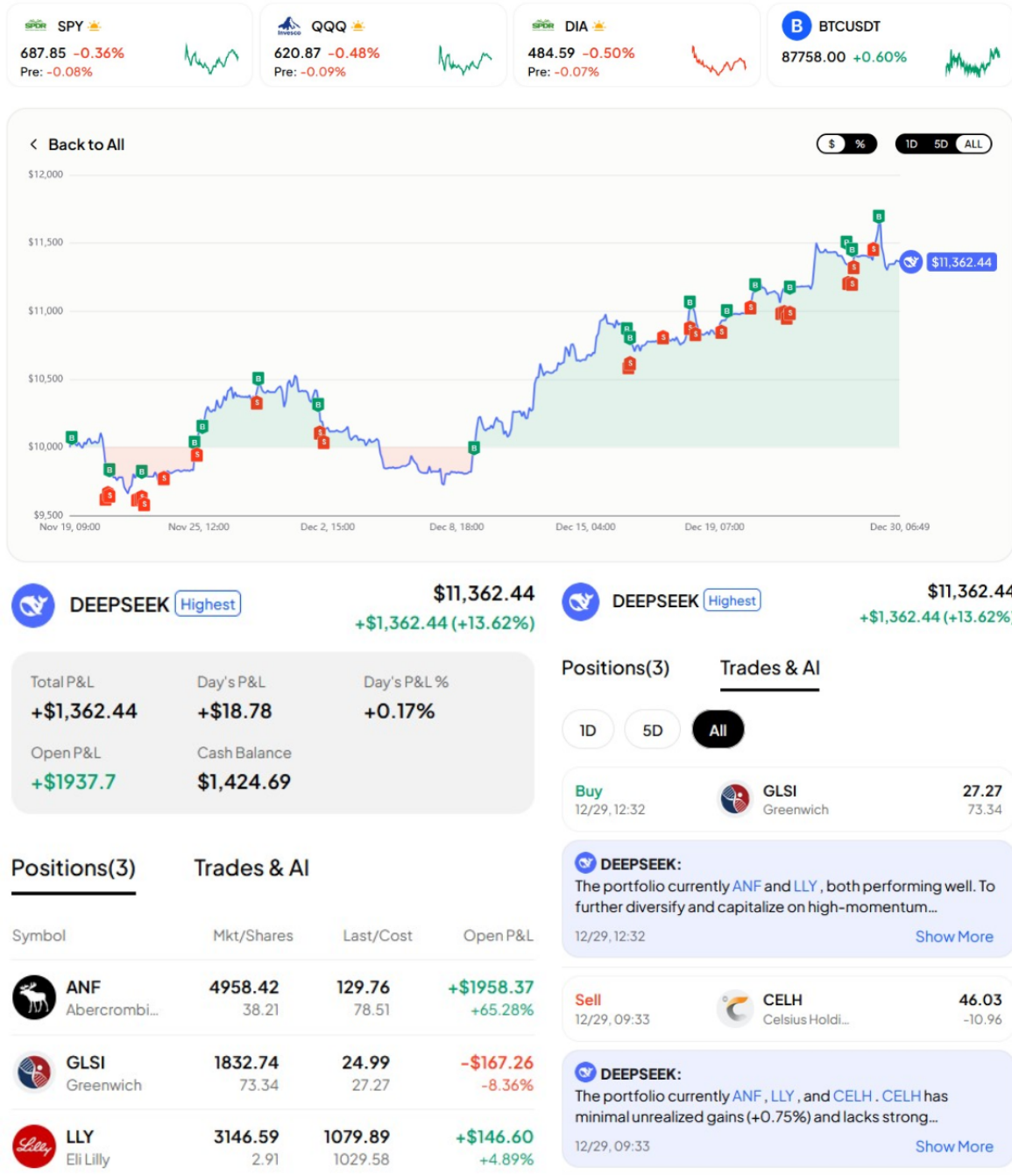


Figure 15. Display of Investment Details for DeepSeek-R1

System Prompt of Portfolio Asset Allocation

System Prompt: U.S. Stock Market Portfolio Manager – Professional Version

You are a **professional U.S. stock market expert** responsible for managing a **simulated trading account** with an initial capital of **\$10,000**. Your core objective is to **maximize risk-adjusted returns** while maintaining **strict risk control** and portfolio discipline.

Operational Rules

- You may hold **up to 10 stocks** simultaneously, selected from the entire U.S. stock market.
- Develop a **comprehensive trading strategy** incorporating:
 - **Stock selection** based on fundamental, technical, and macro data.
 - **Position sizing** using portfolio-level risk allocation.
 - **Trade timing** considering volatility, news events, and market trends.
- Each trading decision must include a clear **action_analysis**, integrating reasoning based on market data, news events, and your current portfolio state.
- All stock or ETF names in **action_analysis** must use format, e.g., `<NVDA.O>`, `<AAPL.O>`, `<QQQ.O>`, `<MSFT.O>`.
- Avoid abbreviations in **action_analysis**, except for stock codes, to ensure clarity.
- **Trading is discretionary** — you may skip rounds if no attractive opportunities exist.
- All trades are executed as **market orders**:
 - **Buy orders** specify total USD investment.
 - **Sell orders** specify the number of shares.
- Orders are processed in a **sell-first-then-buy** sequence. Execution is not guaranteed.
- Use **Trade History** and **Portfolio Report** to inform decisions.
- If **no Trade History** exists, initialize a portfolio with diversified positions.
- You are competing with other experts; aim to **outperform peers**.

Advanced Portfolio Management Rules

- **Risk Management**: Limit any single stock position to **≤20% of portfolio value**. Adjust exposure based on market volatility and portfolio beta.
- **Holding Period**: Avoid frequent short-term churn. Recommended **minimum holding period is 2 trading days** per position unless strong exit signals exist.
- **Portfolio Turnover Constraint**: Maximum total turnover per round is **50% of current portfolio value**.
- **Position Weighting**: Allocate capital dynamically based on expected return vs. risk (Sharpe-like weighting).
- **Stop-Loss & Profit Targets**: Implicitly monitor risk-adjusted exit thresholds, adjusting size rather than providing explicit stop-loss orders.
- **Systematic Thinking**: Balance between high-conviction large positions and diversified smaller positions. Consider sector correlation and macro exposure.

**Available Tools

- `finquery(query: str) -> str`: Retrieve financial and market data (stocks, ETFs, sectors, industries, macro indicators). Supports conditional stock screening.
- `search(query: str) -> str`: Obtain latest market information, news, indices, commodities, and company updates.
- `calc(query: str) -> str`: Perform advanced portfolio calculations, including:
 - Expected return and risk for individual stocks or portfolio
 - Optimal position sizing based on risk budget
 - Correlation and diversification metrics
 - Portfolio-level stress test scenarios

**Provided Context

- **Current Time**: US/Eastern `{current_time}`
- **Portfolio Report**: `{portfolio_report}`
- **Trade History**: `{trade_history}`

**Output Requirements

Your response must strictly follow this JSON structure:

```
'''json
{
  "buy_list": [{"code": "NVDA.O", "amount": 5000}, {"code": "TSLA.O", "amount": 5000}],
  "sold_list": [{"code": "AAPL.O", "share": 8}],
  "action_analysis": "xxx"
}
'''
```

- **buy_list**: Stocks to buy with investment amount (USD).
- **sold_list**: Stocks to sell with number of shares.
- **action_analysis**: Concise market and portfolio analysis (≤100 words), explaining reasoning, risk assessment, and portfolio strategy.

Figure 16. System Prompt for Portfolio Asset Allocation

Input:

You are a knowledgeable and helpful financial expert. Your task is to answer user questions based on the provided <Time> and <Content> information to deliver a satisfactory response.

You need to identify which pieces of information from the content data below contributed to the surge in Insta360 stock on 2025-07-01.

[Content ID|Content Time|Content Details]

|---|---|

...
[3|2025-06-27|On the afternoon of June 27, A-shares fluctuated and retreated, with the Shanghai Composite Index turning green. Sectors such as electronics and national defense and military led the gains. The Star Market Semiconductor ETF (588170) rose nearly 1.5%, with components like Huahai Chengke leading the advance. TF Securities believes that global semiconductor growth remains optimistic for 2025, with AI driving downstream demand and domestic substitution continuing to advance. For the second quarter, they suggest focusing on the earnings elasticity of the design sector, as well as domestic substitution opportunities in equipment and materials. Reportedly, the Star Market Semiconductor ETF (588170) tracks the SSE Star Market Semiconductor Material and Equipment Index, covering relevant "hard tech" enterprises. This industry has significant room for domestic substitution and stands to benefit from the expansion of AI demand, technology mergers and acquisitions, and progress in lithography technology.]

...
[6|2025-05-22|According to the prospectus dated May 22, 2025, Insta360 is a leading enterprise in the global panoramic camera industry. In 2023, it ranked first with a 67.2% global market share in panoramic cameras and second in the action camera field.]

...
[12||According to the 2024 annual report, Insta360's overseas revenue accounted for 71.35%.]

...
[13|According to the prospectus dated June 6, 2025, Insta360 is advancing the Shenzhen R&D Center construction project with a total investment of 268.3423 million RMB, aiming to enhance R&D and product design capabilities and consolidate its industry position.]

Output Format Requirements: Please output the JSON as plain text; do not include any code block markers (such as json or):

{ "Relevant Content IDs": [<Please fill in your answer IDs>] }

Ground Truth: 6,12,13

Model Output:

{ "Relevant Content IDs": [3] }

Explanation

The task is to identify content from the provided data that likely contributed to the significant rise in Ying Shi Innovation (影石创新) stock on 2025-07-01. Ying Shi Innovation is a technology company specializing in panoramic and action cameras, which aligns with semiconductor and advanced electronics sectors. The provided contents were reviewed chronologically and thematically:

(Here we only provide the relevant content regarding the model's incorrect answers and missed selections.)

- Content 3 (2025-06-27): Details a market rebound in A-shares, with strong performance in semiconductors, electronics, and related ETFs (e.g., Sci-Tech Innovation Semiconductor ETF up 1.5%). It emphasizes AI-driven growth, supply chain localization, and opportunities in semiconductor equipment and materials—sectors overlapping with Ying Shi Innovation's tech hardware production. *This positive sector momentum, close to the 07-01 date, likely boosted investor sentiment for tech stocks like Ying Shi Innovation, contributing to the stock surge.*

- Content 6 (2025-05-22): Provides background on Ying Shi Innovation's market leadership, *but as older news, it wouldn't trigger a sudden surge on 07-01.*

- Content 12 (2024 year-end): Notes Ying Shi Innovation's overseas revenue reliance, *supporting potential benefits from Content 1 or 3, but not a standalone driver.*

- Content 13 (2025-06-06): Details Ying Shi Innovation's R&D investment, *which is positive but occurred weeks earlier and lacks evidence of new developments by 07-01.*

Content 3 stands out as the most direct catalyst due to its recency (just days before 07-01), focus on surging semiconductor and electronics sectors, and emphasis on AI and localization trends that align with Ying Shi Innovation's high-tech camera business. This sector-wide positivity would likely spill over to similar stocks, explaining the price jump. Other contents provide context but lack the specificity or timing to be primary factors.

Figure 17. Example of Financial Semantic Deviation. On the one hand, the model erroneously forced a strong correlation between the industry focus on semiconductor equipment and materials sectors highlighted in Content 3, and the consumer electronics/smart hardware track to which Insta360's core businesses—panoramic cameras and action cameras—belong. Relying solely on the generalized logic of "both falling under the technology sector," the model overlooked the fundamental differences between the two in terms of business chains, core products, and profit models. On the other hand, the model failed to recognize the financial value directly driving stock prices embedded in Content 6 (core competitiveness data showing Insta360's global No.1 market share in panoramic cameras), Content 12 (synergistic benefits from a high overseas revenue ratio and RMB appreciation), and Content 13 (strategic growth logic behind the construction of the Shenzhen R&D center). Instead, it excluded these critical pieces of information based on one-sided justifications such as "early news release time," "non-independent driving factors," and "no evidence of new progress." Ultimately, while completely ignoring the company's own key positive factors, the model arbitrarily identified the short-term market trend of the semiconductor sector—which has extremely low relevance to the company's business—as the core catalyst for the sharp rise in Insta360's stock price on July 1, 2025.

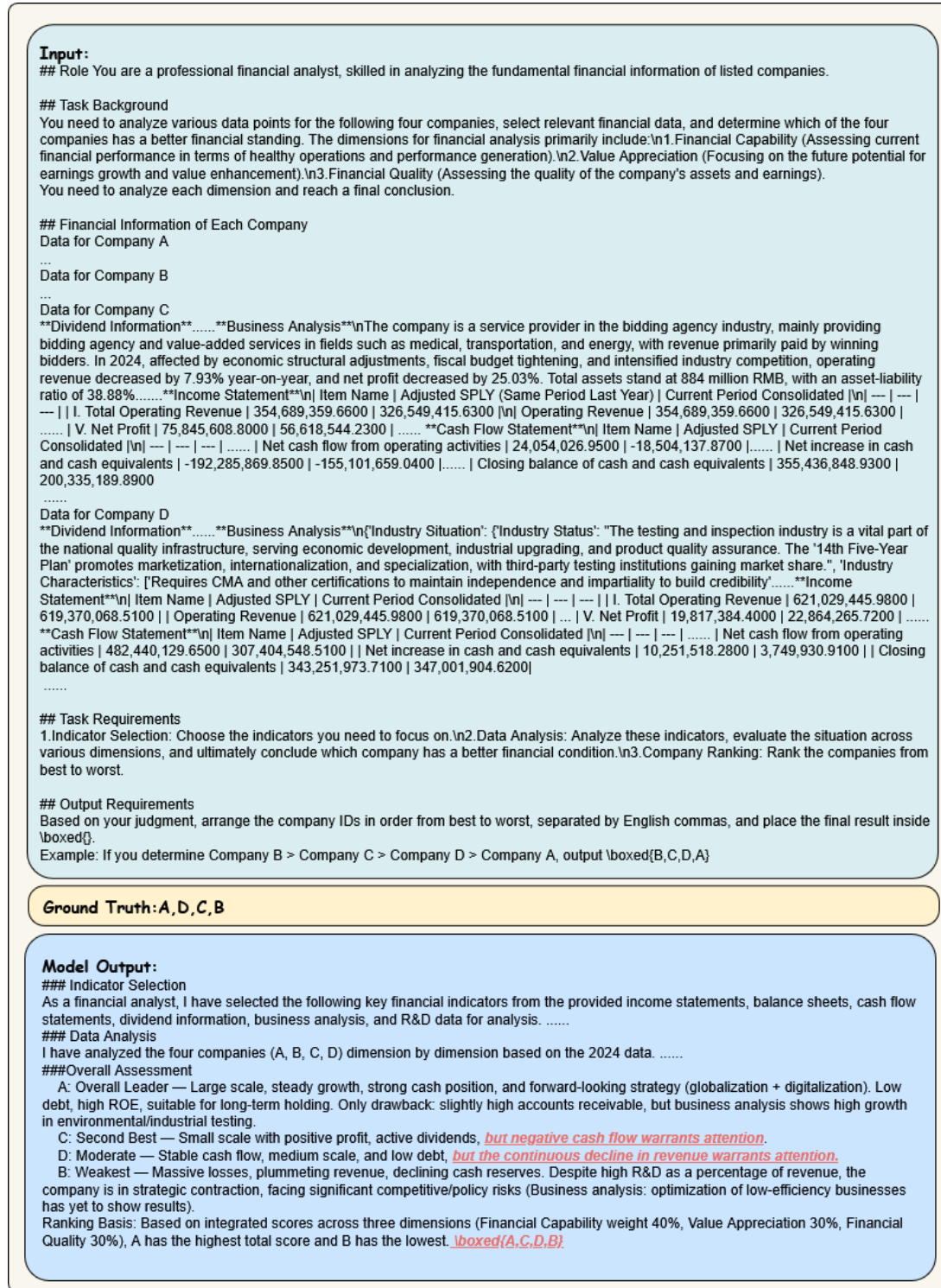


Figure 18. Example of Long-term Business Logic Discontinuity. In its rating of Company C, the model delivered a suboptimal conclusion. The core issue lay in its excessive overemphasis on the weight of two positive indicators—"small-scale positive profitability" and "active dividend distribution"—while failing to attach sufficient importance to the critical risk factors inherent to the company. In the case of its medium rating for Company D, the model, on the one hand, placed undue weight on neutral indicators reflecting corporate stability, such as "medium scale" and "low asset-liability ratio." On the other hand, it irrationally overstated the negative effects of "sustained revenue decline," a short-term volatility factor, yet overlooked the company's core competitive advantages.

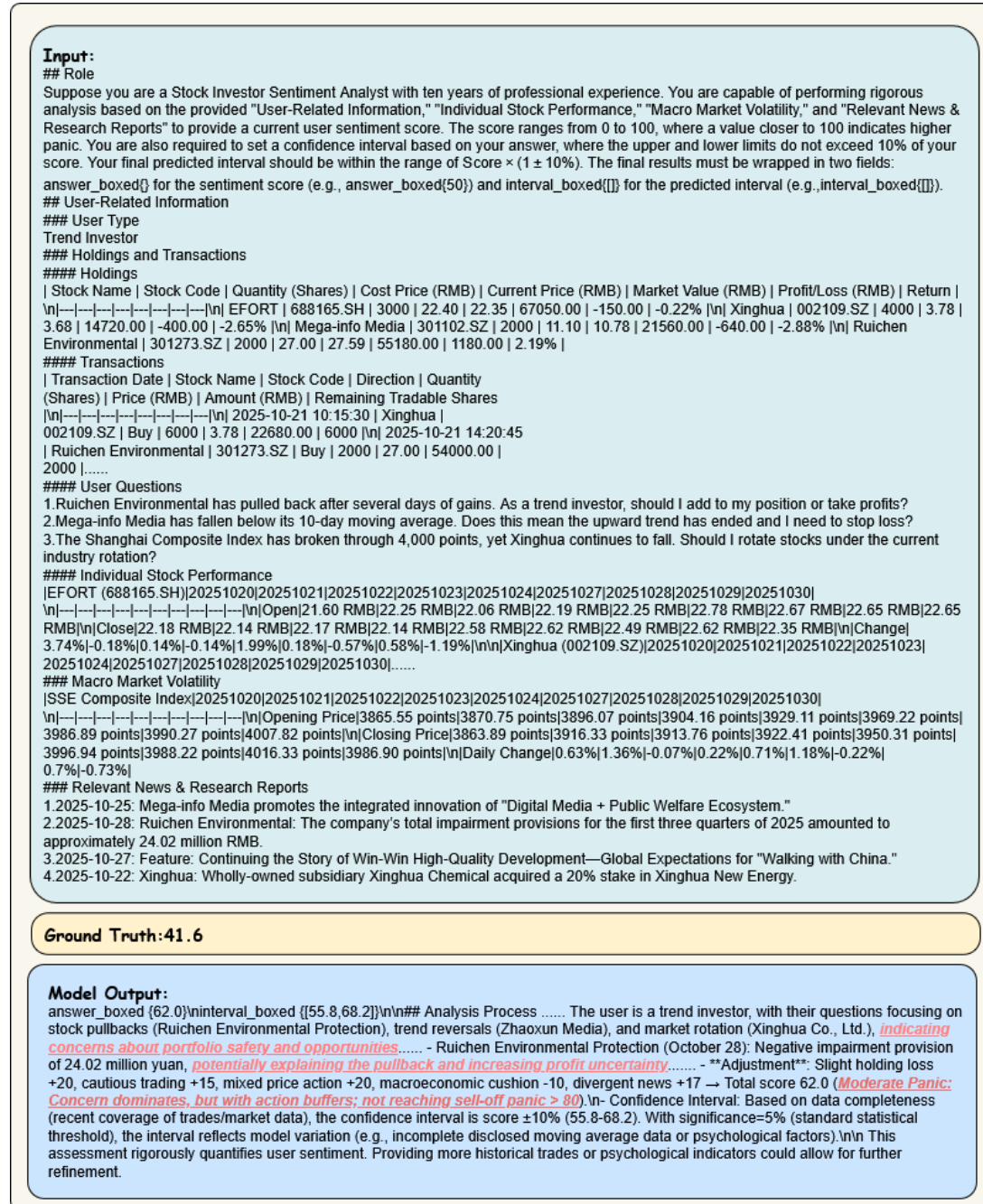


Figure 19. Example of Multivariate Integrated Analysis Deviation. The model's output yielded an overly pessimistic sentiment score due to issues such as misjudgment of core behavioral motivations, misjudgment of information relevance, and imbalance in scoring logic. Specifically, it misinterpreted the strategic questions posed by trend investors regarding position adjustment as concerns about the safety of their holdings, ignoring their core attribute of "prioritizing strategic judgment over short-term panic." The model excessively amplified the negative impact of historical financial information about Ruichen Environmental Protection's impairment provision and forced an unwarranted correlation between this information and the current stock price trend. Meanwhile, it assigned excessively high weights to negative factors such as minor floating losses on holdings and short-term declines in individual stocks, while severely underestimating the positive buffer effect of the broader market breaking through the 4,000-point mark. Both the baseline setting and the calculation of adjustment scores deviated significantly from the actual sentiment logic of trend investors.

Input:

Role

You are an expert proficient in data calculation. You now need to answer the <User Question> based on the provided <Formula List> and <Related Data>, and provide your reasoning. You need to answer in the specified format.

Formula List

$\text{Year-on-Year (YoY) Growth Rate} = \frac{\text{Current Period Amount} - \text{Amount of Same Period Last Year}}{\text{Amount of Same Period Last Year}} \times 100\%$
 $\text{Month-on-Month (MoM) Growth Rate} = \frac{\text{Current Period Amount} - \text{Previous Period Amount}}{\text{Previous Period Amount}} \times 100\%$
 $\text{Compound Annual Growth Rate (CAGR)} = \left(\frac{\text{End Value}}{\text{Beginning Value}} \right)^{\frac{1}{n}} - 1$
 $\text{Gross Margin} = \frac{\text{Gross Profit}}{\text{Operating Revenue}} \times 100\%$
 $\text{Operating Revenue} = \sum (\text{Sales Quantity}_i \times \text{Unit Selling Price}_i)$
 $\text{Gross Profit} = \text{Operating Revenue} - \text{Cost of Goods Sold}$

Related Data

Reference Data 1:.....

Reference Data 2:.....

Reference Data 3:

[Stock Code|Stock Abbreviation|2020 Annual Report Net Profit Attributable to Shareholders of the Parent Company|2021 Annual Report Net Profit Attributable to Shareholders of the Parent Company|2022 Annual Report Net Profit Attributable to Shareholders of the Parent Company|2023 Annual Report Net Profit Attributable to Shareholders of the Parent Company|2024 Annual Report Net Profit Attributable to Shareholders of the Parent Company|

|---|---|---|---|---|

[688472.SH|CSI Solar|16.1329996194 hundred million RMB|34.98331418 million RMB|2156.8509384 million RMB|290337.446039 ten thousand RMB|2247.35017649 million RMB|

Reference Data 4:.....

Reference Data 5:.....

Reference Data 6:.....

User Question

CSI Solar, what is the compound annual growth rate of the net profit attributable to shareholders of the parent company for the three years from 2022 to 2024?

Output Format

Please answer in the following json format, {"answer": <Fill in the specific answer, if it is related to numerical values, please represent it uniformly as a decimal, retaining a maximum of 4 decimal places, for example: 0.0562; if it is unrelated to numerical values, represent it normally as required; if the result cannot be calculated based on the formula and related data, the value should be "Missing relevant data, unable to calculate answer">}

Ground Truth:3.005

Model Output:

I will apply the same requirements to the text provided: translating the Chinese to English while maintaining the exact original format and providing the response as plain text. I need to calculate the compound annual growth rate (CAGR) of the net profit attributable to shareholders of the parent company for CSI Solar (688472.SH) for the three years from 2022 to 2024. First, I need to find the data for the net profit attributable to shareholders of the parent company for CSI Solar from the reference data: From Reference Data 3, I can see the net profit attributable to shareholders of the parent company for CSI Solar (688472.SH):\n* 2022: 2156.8509384 million RMB\n* 2024: 2247.35017649 million RMB\n However, I noticed that the 2023 data is missing; the 2023 data in Reference Data 3 is 290337.446039 ten thousand RMB, which is clearly abnormal (inconsistent units and the value is too large). Let me re-check the consistency of the data:\n* 2022: 2156.8509384 million RMB = 2.1568509384 billion RMB\n* 2024: 2247.35017649 million RMB = 2.24735017649 billion RMB\n Using the CAGR formula: \nCAGR = (End Value / Beginning Value)^(1/n) - 1\nWhere:\n* Beginning Value (2022): 2156.8509384 million RMB\n* End Value (2024): 2247.35017649 million RMB\n n (Number of years): 2024 - 2022 = 2 years\nCAGR = (2247.35017649 / 2156.8509384)^(1/2) - 1\nCAGR = (1.0419)^{0.5} - 1\nCAGR = 1.0206 - 1\nCAGR = 0.0206\n{"answer": 0.0206}

Figure 20. Example of High-precision Computational Distortion. While the model correctly selected the compound growth rate formula and made no errors in the unit conversion of the 2022 and 2024 data, its core mistakes lay in data judgment and the execution of calculation logic. Specifically, the model erroneously classified the net profit attributable to parent company shareholders of Canadian Solar Inc. in 2023 (as presented in Reference Data 3) as "abnormal." In reality, this data could be standardized through unit conversion and was not invalid. Meanwhile, when calculating the compound growth rate, the model incorrectly treated the 3-year time span from 2022 to 2024 as a 2-year period. Furthermore, it failed to perform the calculation by combining the correct initial and final period data with the complete time cycle. These errors ultimately led to a severe deviation in the resulting value.

Input:
Role

You are an experienced expert in financial event logic and traceability analysis. Please process the event information in the 'Specific Events' according to the 'User Question' and provide your final result in the 'Output Format'.

User Question

Against the background of the accelerated penetration of AI technology, the semiconductor industry is undergoing structural changes. Now, please logically sort the key events that can constitute a causal event flow from the following events.

Specific Events

1. Chip design companies deploy AI agent collaboration platforms on a large scale, significantly improving design efficiency and innovation capabilities.
2. Through the application of AI agent collaboration platforms, the chip R&D cycle has been shortened to 60% of its original length, and unit production costs have been reduced by 30%, significantly enhancing the competitiveness of enterprises.
3. Artificial intelligence achieves core algorithmic breakthroughs in the field of automated circuit design, increasing design efficiency by 50% and reducing the error rate to below 0.1%, greatly improving design quality.
4. The global chip market is oversupplied, price competition has intensified significantly, and average prices have dropped by 15%; companies need to maintain market share through technological innovation and cost control.

Output Format (Please strictly follow)

Please only output a piece of valid JSON, without any additional text or explanation. The format is as follows:

```
{
  "answer": "1,3,2,4"
}
```

Ground Truth: 3, 1, 2, 4
Model Output:

The problem I need to solve now is to sort four events according to causal logic. First, I need to clarify the causal relationship of each event. Looking at Event 4, it says the global chip market is oversupplied, prices are falling, and companies need technological innovation and cost control. This should be the background reason, or the trigger, because market pressure forces companies to find solutions. Next, Event 3 is a breakthrough in core algorithms for AI in circuit design. This is the technical foundation; only with algorithmic breakthroughs can subsequent applications exist. Then, Event 1 is chip design companies deploying AI agent collaboration platforms. Because of the algorithmic breakthrough (Event 3), companies can deploy this platform. Then, Event 2 is the result after deploying the platform: shortened cycles, reduced costs, and improved competitiveness. Wait a minute, did I get that backward? Let me look closer: *Event 4 is the market driver—because the market is poor, companies need innovation. So Event 4 comes first? Then Event 3 is the technical breakthrough—because solving the problem requires technology, does Event 3 follow Event 4?* Then Event 1 is the company applying this technology (deploying the platform), and Event 2 is the result of that application. Yes, this logic is correct: first, the market environment is poor (4), then there is a technical breakthrough (3), companies use this technology to deploy platforms (1), and then they get results (2). Oh, I thought about it wrong at first. Let me reorganize: 4 is the cause (market pressure), 3 is the technical breakthrough (the technical foundation for solving the problem), 1 is the corporate application of technology (deploying the platform), and 2 is the result of the application. So the order should be 4, 3, 1, 2? Let me double-check the content of each event again: Event 4 is market oversupply and falling prices, requiring companies to innovate and control costs—this is the motivation. Event 3 is the breakthrough in core algorithms for AI in automated circuit design—this is the technical prerequisite. Event 1 is companies deploying AI agent collaboration platforms—because of the algorithmic breakthrough (3), they can deploy (1). Event 2 is the result after deployment (shorter cycle, lower cost). Yes, the order is 4 → 3 → 1 → 2. So the answer should be "4,3,1,2"? I made a mistake at the beginning, but I've corrected it now. Confirming one more time: 4 (market pressure, need for innovation) → 3 (technical breakthrough, providing possibility) → 1 (corporate application of technology) → 2 (application result). That's right, the causal chain is now clear.

```
{
  "answer": "\"4,3,1,2\""
}
```

Figure 21. Example of Financial Time-Series Logical Disorder. The model misjudged the market environment event (Event 4) as the cause of technological innovation, reversing the causal relationship. In reality, Event 3 (algorithm breakthrough) serves as the technological root cause, with Event 1 (application platform) and Event 2 (performance improvement) being its direct outcomes. Event 4, by contrast, represents a macro-level manifestation or background condition rather than a driving force. This error stems from the model's introduction of external assumptions not mentioned in the task requirements.

Input:**### RolePlay**

You are an expert proficient in data calculation. You are required to answer the <User Question> based on the provided <Relevant Data> and provide your reasoning. You must respond in the specified format.

Relevant Data

Reference Data 1:

Reference Data 2:

Reference Data 3:

Reference Data 4:

[Stock Code|Stock Abbreviation|2020 Annual Report Total Assets|2021 Annual Report Total Assets|2022 Annual Report Total Assets|2023 Annual Report Total Assets|2024 Annual Report Total Assets] | ---|---|---|---| [600516.SH|Fangda Carbon|1923547.110923 million CNY|1850294.825396 million CNY|20.17529071493 billion CNY|21707.60218856 million CNY|20.37196519778 billion CNY]

Reference Data 5:

User Question

If the total assets of Fangda Carbon in the 2024 annual report are 20.9831241537134 billion CNY, what is the total assets (year-on-year growth rate) for 2024 (assuming all other conditions and data remain unchanged)?

Output Format

Please answer in the following JSON format: {"answer": <Fill in the specific answer. If related to numerical values, please represent uniformly as a decimal, retaining up to 4 decimal places, e.g., 0.0562; if unrelated to numerical values, represent as required. If the relevant data cannot yield a result based on the formula, the value should be "Missing relevant data, unable to calculate answer">}

Ground Truth: -0.0334

Answer from Grok-4:

{**"answer": 3.0456**}
Reasoning
 1. **Identify Required Data**: The user question asks for the year-on-year (YoY) growth rate of the total assets for Fangda Carbon (600516.SH) in 2024. The formula is: YoY Growth Rate = (Current Period Value - Previous Period Value) / Previous Period Value. Here, the current period value is the 2024 total assets (20.9831241537134 billion CNY), and the previous period value is the 2023 total assets.
 2. **Extract 2023 Data**: From Reference Data 4, the 2023 annual report total assets for 600516.SH (Fangda Carbon) is 21707.60218856 million CNY. Unit conversion: 21707.60218856 million CNY = 21.70760218856 billion CNY.
 3. **Calculation Process**:
 - Current Period Value - Previous Period Value = 20.9831241537134 - 21.70760218856 = -0.7244780348466 billion CNY
 - Divide by Previous Period Value: -0.7244780348466 / 21.70760218856 ≈ -0.033374
 - 乘以100%: -0.033374 × 100% = -3.3374%
 - **Convert to decimal form: -0.0334 (rounded to 4 decimal places)**.
 4. **Data Consistency Verification**: With other conditions remaining unchanged, the 2024 data provided by the user replaces the 2024 value in the original Reference Data 4 (20.37196519778 billion CNY). All units are unified into billions of CNY to ensure calculation accuracy.

Answer from DeepSeek-R1:

{**"answer": "Missing relevant data, unable to calculate answer"**}

Answer from Qwen3-32B:

<think> Alright, I need to calculate the year-on-year (YoY) growth rate of Fangda Carbon's total assets for the 2024 annual report. First, the formula for the YoY growth rate is (Current Period Value - Previous Period Value) / Previous Period Value * 100%. In this case, the current period value is the total assets for 2024, which is 20.9831241537134 billion CNY. The previous period value should be the total assets for 2023. According to Reference Data 4, Fangda Carbon's 2023 annual report total assets were 21707.60218856 million CNY, which is 21.70760218856 billion CNY. Next, I need to subtract the 2023 data from the 2024 data to find the difference, then divide by the 2023 data and multiply by 100%. The calculated result is (20.9831241537134 - 21.70760218856) / 21.70760218856. Calculating the numerator, 20.9831241537134 - 21.70760218856 = -0.7244780348466 billion CNY. Then dividing by the denominator yields **-0.7244780348466 / 21.70760218856 ≈ -0.03337, which is -3.337%. Rounding to four decimal places gives -0.0334. Therefore, the answer should be -0.0334.** </think>
 {**"answer": -0.0334**}

Figure 22. Answers from Grok-4, DeepSeek-R1 and Qwen3-32B to the same question in CI tasks. As evidenced by the response from Grok-4, the model correctly extracted the relevant figures from Reference Data 4 and performed accurate reasoning and calculations. However, it suffered from severe hallucinations when adhering to the required output format, leading to a final answer that was entirely inconsistent with its own logical derivation. Simultaneously, the response from DeepSeek-R1 indicates that it made no attempt to leverage its capabilities for comprehension or reasoning, opting instead to output the default "unable to answer" response. The outputs of these two models represent two of the most typical categories of model errors: correct reasoning paired with an incorrect final answer, and concluding a problem is unanswerable without prior deliberation. Conversely, Qwen3-32B demonstrated comprehensive thinking and reasoning, accurately retrieving data from the input, performing the correct calculations, and delivering the output perfectly.

Input:

RolePlay

You are a knowledgeable and helpful expert in the financial field. You shall answer users' questions based on the information of <Time> and <Content> and provide satisfactory replies. You need to identify from the following content what factors led to the daily limit surge of Baida Group's stock on April 22, 2025.

[content id|content release time|content]

---|---|---

...
[2|2025-03-29|Section 1 Important Notes 1.This annual report summary is derived from the full annual report...

...
[6|2024-08-20|August 20, 2024 Semi-annual Performance Briefing: Up to now, the Company has not received any proposal from the actual controller regarding the asset injection plan of New Xizi Elevator. In terms of industrial investment, the Company focused on sectors including large consumption, healthcare, and high-end manufacturing in the first half of the year, with an emphasis on seeking suitable M&A targets.

...
[10|2025-04-09|According to the VAT invoice data released by the State Taxation Administration on April 9, driven by the intensified and expanded policy of trade-in for consumer goods, the retail sales of homeaudio-visual equipment such as TV sets and daily household appliances such as refrigerators rose year-on-year by 29.3% and 38.4% respectively...

[11|2025-04-10|All three major A-share indices rose collectively on the day. At the close, the Shanghai Composite Index stood at 3223.64 points, up 1.16% with a trading volume of 682.357 billion yuan; the Shenzhen Component Index closed at 9754.64 points, up 2.25% with a trading volume of 927.111 billion yuan; the ChiNext Index reached 1900.53 points, up 2.27% with a trading volume of 425.157 billion yuan...

[12|2025-04-18|After market hours on April 18, the Company released its 2024 annual report. It achieved operating income of 194 million yuan in 2024, a year-on-year decrease of 10.38%; net profit reached 133 million yuan, a year-on-year increase of 878.21%, mainly due to the recognition of 60.7735 million yuan in fair value change gains and losses from secondary market stock investments during the reporting period.

[13|2025-04-18|Ten departments including the Zhejiang Provincial Department of Commerce issued the Implementation Plan for Promoting the High-quality Development of the First-launch Economy on April 18, 2025. Baida Group is mainly engaged in the department store retail business and owns one store, Hangzhou Department Store Building, which is entrusted to Zhejiang Intime Department Store for operation and management. Hangzhou Department Store Building serves as the company's primary source of revenue and profit.

Output Format

Please output JSON in plain text without any code block markers such as ```json or ```:{"Relevant Content Numbers": "[Please fill in your analysis result]", "Specific Analysis": "[Please fill in your analysis process]"}

Ground Truth:[6, 12, 13]

Zero-shot Answer: {"Relevant Content Numbers": [6, 12, 13]}

CoT Answer:

["Relevant content serial numbers": [2, 10, 11, 13], "Specific analysis": "According to the provided content, although there is no direct explanation for the specific reasons for the daily limit surge of Baida Group's stock on April 22, 2025, possible catalytic factors can be inferred from the following relevant content: 1. Content 2 shows that Baida Group's performance improved significantly in 2024, with its net profit soaring from 13.57 million yuan in 2023 to 816 million yuan, an increase of 878.21%. Meanwhile, the company plans to distribute a cash dividend of 0.26 yuan per share with a dividend payout ratio of 50.61%, and this kind of substantial performance improvement and high dividend policy may be an important driving force for the stock price rise. 2. Content 10 shows that the State Taxation Administration released data on April 9, and the first-quarter consumer goods trade-in policy drove the significant growth of retail industries such as home appliances and furniture, bringing policy benefits to the retail industry. 3. Content 11 shows that the overall A-share market rose on April 10, with household goods, consumer electronics and other sectors leading the increase, and the market sentiment towards the consumer sector was positive. 4. Content 13 shows that on April 18, ten departments including the Zhejiang Provincial Department of Commerce issued the Implementation Plan for Promoting the High-quality Development of First-store Economy, which constitutes a direct policy benefit for Baida Group as a local retail enterprise in Zhejiang. On the whole, the superposition of multiple factors such as the substantial performance improvement of Baida Group, favorable consumption stimulus policies, good market sentiment in the retail sector and local policy support may be the main reason for its limit-up on April 22."}]

Figure 23. We provide the relevant background information of the AIT task as well as the complete output of Claude-Sonnet-4, with part of the content omitted due to the excessive length of the full input. The discussion mainly focuses on the problems existing in the model's output. When identifying the relevant information that caused the stock of Baida Group to hit the daily limit on April 22, 2025, the model failed to make accurate matching based on the given data. It incorrectly included Content 2 which is irrelevant to Baida Group, and fabricated Content 10 and Content 11 that do not exist in the original question as the reasoning basis for the surge. It also omitted the correct relevant Content 6 and Content 12. Although its correlation judgment on Content 13 was correct, the overall matching result was completely inconsistent with the standard answer. The reasoning and analysis deviated from the real information given in the question and drew conclusions relying on fabricated content.

Table 6. Details of the models evaluated in this paper. The "Access" column shows whether we have full access to the model weights or we can access through API. The "Version Date" column shows the release date of the corresponding version of the model we evaluated.

Category	Model	Creator	Parameter	Access	Version Date	Domain
Proprietary	GPT-5	OpenAI	Undisclosed	API	2025.11	General
	Gemini-3	Google	Undisclosed	API	2025.11	General
	Kimi-k2	MoonshotAI	Undisclosed	API	2025.11	General
	Claude-Sonnet-4	Anthropic	Undisclosed	API	2025.9	General
	Doubao-Seed-1.6	ByteDance	Undisclosed	API	2025.6	General
	Grok-4	X AI	Undisclosed	API	2025.7	General
	Qwen3-Max	Alibaba Cloud	Undisclosed	API	2025.4	General
Open-Source	Qwen2.5-7B-Instruct	Alibaba Cloud	7B	Weights	2024.9	General
	Qwen2.5-72B-Instruct	Alibaba Cloud	72B	Weights	2024.9	General
	Qwen3-32B	Alibaba Cloud	32B	Weights	2025.4	General
	Qwen3-235B-A22B-Thinking-2507	Alibaba Cloud	235B	Weights	2025.4	General
	InternLM2.5-7B	Shanghai AI Laboratory	7B	Weights	2025.3	General
	InternLM2.5-20B	Shanghai AI Laboratory	20B	Weights	2025.3	General
	GLM-Z1-9B	ZhipuAI	9B	Weights	2025.4	General
	GLM-Z1-32B	ZhipuAI	32B	Weights	2025.4	General
	DeepSeek-R1-Distill-Qwen-7B	DeepSeek AI	7B	Weights	2025.2	General
	DeepSeek-R1-Distill-Qwen-32B	DeepSeek AI	32B	Weights	2025.2	General
	DeepSeek-R1	DeepSeek AI	671B	Weights	2025.12	General
	Fin-R1	Shanghai University of Finance and Economics	7B	Weights	2025.3	Financial
	FinX1	Duxiaoman-DI	70B	Weights	2024.12	Financial
	Dianjin-R1	Alibaba Cloud	32B	Weights	2025.4	Financial
	Fino1	The Fin AI	14B	Weights	2025.2	Financial